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Method and apparatus for beam measurement and reporting

Abstract

A method for operating a user equipment (UE) comprises receiving information for a maximum permissible emission (MPE) report; determining the MPE report including Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$; including the MPE report in a power headroom (PHR) report; and transmitting, via medium access control-control element (MAC CE), the PHR report including the MPE report.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS AND CLAIM OF PRIORITY (1) The present application claims priority to U.S. Provisional Patent Application No. 63/190,111, filed on May 18, 2021; U.S. Provisional Patent Application No. 63/215,860, filed on Jun. 28, 2021; U.S. Provisional Patent Application No. 63/217,024, filed on Jun. 30, 2021; U.S. Provisional Patent Application No. 63/218,169, filed on Jul. 2, 2021; and U.S. Provisional Patent Application No. 63/236,075, filed on Aug. 23, 2021. The content of the above-identified patent documents is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to wireless communication systems and more specifically to a method and apparatus for beam measurement and reporting.

BACKGROUND

(2) Understanding and correctly estimating the channel between a user equipment (UE) and a base station (BS) (e.g., gNode B (gNB)) is important for efficient and effective wireless communication. In order to correctly estimate the DL channel conditions, the gNB may transmit a reference signal, e.g., CSI-RS, to the UE for DL channel measurement, and the UE may report (e.g., feedback) information about channel measurement, e.g., CSI, to the gNB. With this DL channel measurement, the gNB is able to select appropriate communication parameters to efficiently and effectively perform wireless data communication with the UE.

SUMMARY

(3) Embodiments of the present disclosure provide methods and apparatuses for beam measurement and reporting.

(4) In one embodiment, a UE for CSI reporting in a wireless communication system is provided. The UE includes a transceiver configured to receive information for a maximum permissible emission (MPE) report. The UE further includes a processor operably connected to the transceiver. The processor, based on the information, is configured to: determine the MPE report including Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$, and include the MPE report in a power headroom (PHR) report. The transceiver is further configured to transmit via medium access control control element (MAC CE) the PHR report including the MPE report.

(5) In another embodiment, a BS in a wireless communication system is provided. The BS includes a processor configured to generate information for a MPE report. The BS further includes a transceiver operably connected to the processor. The transceiver is configured to: transmit the information; and receive, via a MAC CE, a PHR report, the PHR report including the MPE report, wherein the MPE report

includes Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$.

(6) In yet another embodiment, a method for operating a UE is provided. The method comprises: receiving information for a MPE report; determining the MPE report including Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$; including the MPE report in a PHR report; and transmitting, via a MAC CE, the PHR report including the MPE report.

(7) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(8) Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “couple” and its derivatives refer to any direct or indirect communication between two or more elements, whether or not those elements are in physical contact with one another. The terms “transmit,” “receive,” and “communicate,” as well as derivatives thereof, encompass both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The term “controller” means any device, system or part thereof that controls at least one operation. Such a controller may be implemented in hardware or a combination of hardware and software and/or firmware. The functionality associated with any particular controller may be centralized or distributed, whether locally or remotely. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

(9) Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

(10) Definitions for other certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

(2) FIG. 1 illustrates an example wireless network according to embodiments of the present disclosure;

(3) FIG. 2 illustrates an example gNB according to embodiments of the present disclosure;

(4) FIG. 3 illustrates an example UE according to embodiments of the present disclosure;

(5) FIG. 4A illustrates a high-level diagram of an orthogonal frequency division multiple access transmit path according to embodiments of the present disclosure;

(6) FIG. 4B illustrates a high-level diagram of an orthogonal frequency division multiple access receive path according to embodiments of the present disclosure;

(7) FIG. 5 illustrates a transmitter block diagram for a PDSCH in a subframe according to embodiments of the present disclosure;

(8) FIG. 6 illustrates a receiver block diagram for a PDSCH in a subframe according to embodiments of the present disclosure;

(9) FIG. 7 illustrates a transmitter block diagram for a PUSCH in a subframe according to embodiments of the present disclosure;

(10) FIG. 8 illustrates a receiver block diagram for a PUSCH in a subframe according to embodiments of the present disclosure;

(11) FIG. 9 illustrates an example antenna blocks or arrays forming beams according to embodiments of the present disclosure;

(12) FIG. 10 illustrates an uplink multi-beam operation according to embodiments of the present disclosure;

(13) FIG. 11 illustrates an uplink multi-beam operation according to embodiments of the present disclosure;

(14) FIG. 12 illustrates a downlink multi-beam operation 1200 according to embodiments of the present disclosure;

(15) FIG. 13 illustrates a flow diagram of a UE configured to measure P.sub.1 DL measurement RS resources according to embodiments of the present disclosure;

(16) FIG. 14 illustrates a flow diagram of a UE configured to measure P.sub.1 DL measurement RS resources according to embodiments of the present disclosure;

(17) FIG. 15 illustrates an example of single-entry MAC CE for multiple MPE reports according to embodiments of the present disclosure;

(18) FIG. 16 illustrates another example of single-entry MAC CE for multiple MPE reports according to embodiments of the present disclosure;

(19) FIG. 17 illustrates yet another example of single-entry MAC CE for multiple MPE reports according to embodiments of the present disclosure;

(20) FIG. 18 illustrates a flow chart of a method for operating a UE according to embodiments of the present disclosure; and

(21) FIG. 19 illustrates a flow chart of a method for operating a BS according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(22) FIG. 1 through FIG. 19, discussed below, and the various embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged system or device.

(23) The following documents and standards descriptions are hereby incorporated by reference into the present disclosure as if fully set forth herein: 3GPP TS 36.211 v17.0.0, “E-UTRA, Physical channels and modulation” (herein “REF 1”); 3GPP TS 36.212 v17.0.0, “E-UTRA, Multiplexing and Channel coding” (herein “REF 2”); 3GPP TS 36.213 v17.0.0, “E-UTRA, Physical Layer Procedures” (herein

“REF 3”); 3GPP TS 36.331 v17.0.0, “E-UTRA, Medium Access Control (MAC) protocol specification” (herein “REF 4”); 3GPP TS 36.331 v17.0.0, “E-UTRA, Radio Resource Control (RRC) protocol specification” (herein “REF 5”); 3GPP TS 38.211 v17.0.0, “NR, Physical channels and modulation” (herein “REF 6”); 3GPP TS 38.212 v17.0.0, “E-UTRA, NR, Multiplexing and channel coding” (herein “REF 7”); 3GPP TS 38.213 v17.0.0, “NR, Physical Layer Procedures for Control” (herein “REF 8”); 3GPP TS 38.214 v17.0.0, “NR, Physical Layer Procedures for Data” (herein “REF 9”); 3GPP TS 38.215 v17.0.0, “NR, Physical Layer Measurements” (herein “REF 10”); 3GPP TS 38.321 v17.0.0, “NR, Medium Access Control (MAC) protocol specification” (herein “REF 11”); 3GPP TS 38.331 v17.0.0, and “NR, Radio Resource Control (RRC) Protocol Specification” (herein “REF 12”).

(24) Aspects, features, and advantages of the disclosure are readily apparent from the following detailed description, simply by illustrating a number of particular embodiments and implementations, including the best mode contemplated for carrying out the disclosure. The disclosure is also capable of other and different embodiments, and its several details can be modified in various obvious respects, all without departing from the spirit and scope of the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive. The disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

(25) In the following, for brevity, both FDD and TDD are considered as the duplex method for both DL and UL signaling.

(26) Although exemplary descriptions and embodiments to follow assume orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA), the present disclosure can be extended to other OFDM-based transmission waveforms or multiple access schemes such as filtered OFDM (F-OFDM).

(27) To meet the demand for wireless data traffic having increased since deployment of 4G communication systems, efforts have been made to develop an improved 5G or pre-5G communication system. Therefore, the 5G or pre-5G communication system is also called a “beyond 4G network” or a “post LTE system.”

(28) The 5G communication system is considered to be implemented in higher frequency (mmWave) bands, e.g., 60 GHz bands, so as to accomplish higher data rates or in lower frequency bands, such as below 6 GHz, to enable robust coverage and mobility support. To decrease propagation loss of the radio waves and increase the transmission coverage, the beamforming, massive multiple-input multiple-output (MIMO), full dimensional MIMO (FD-MIMO), array antenna, an analog beam forming, large scale antenna techniques and the like are discussed in 5G communication systems.

(29) In addition, in 5G communication systems, development for system network improvement is under way based on advanced small cells, cloud radio access networks (RANs), ultra-dense networks, device-to-device (D2D) communication, wireless backhaul communication, moving network, cooperative communication, coordinated multi-points (CoMP) transmission and reception, interference mitigation and cancellation and the like.

(30) The discussion of 5G systems and frequency bands associated therewith is for reference as certain embodiments of the present disclosure may be implemented in 5G systems. However, the present disclosure is not limited to 5G systems, or the frequency bands associated therewith, and embodiments of the present disclosure may be utilized in connection with any frequency band. For example, aspects of the present disclosure may also be applied to deployment of 5G communication systems, 6G or even later releases which may use terahertz (THz) bands.

(31) FIGS. 1-4B below describe various embodiments implemented in wireless communications systems and with the use of orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA) communication techniques. The descriptions of FIGS. 1-3 are not meant to imply physical or architectural limitations to the manner in which different embodiments may be implemented. Different embodiments of the present disclosure may be implemented in any suitably-arranged communications system. The present disclosure covers several components which can be used in conjunction or in combination with one another or can operate as standalone schemes.

(32) FIG. 1 illustrates an example wireless network according to embodiments of the present disclosure. The embodiment of the wireless network shown in FIG. 1 is for illustration only. Other embodiments of the wireless network **100** could be used without departing from the scope of this disclosure.

(33) As shown in FIG. 1, the wireless network includes a gNB **101**, a gNB **102**, and a gNB **103**. The gNB **101** communicates with the gNB **102** and the gNB **103**. The gNB **101** also communicates with at least one network **130**, such as the Internet, a proprietary Internet Protocol (IP) network, or other data network.

(34) The gNB **102** provides wireless broadband access to the network **130** for a first plurality of user equipments (UEs) within a coverage area **120** of the gNB **102**. The first plurality of UEs includes a UE **111**, which may be located in a small business; a UE **112**, which may be located in an enterprise (E); a UE **113**, which may be located in a WiFi hotspot (HS); a UE **114**, which may be located in a first residence (R); a UE **115**, which may be located in a second residence (R); and a UE **116**, which may be a mobile device (M), such as a cell phone, a wireless laptop, a wireless PDA, or the like. The gNB **103** provides wireless broadband access to the network **130** for a second plurality of UEs within a coverage area **125** of the gNB **103**. The second plurality of UEs includes the UE **115** and the UE **116**. In some embodiments, one or more of the gNBs **101-103** may communicate with each other and with the UEs **111-116** using 5G, LTE, LTE-A, WiMAX, WiFi, or other wireless communication techniques.

(35) Depending on the network type, the term “base station” or “BS” can refer to any component (or collection of components) configured to provide wireless access to a network, such as transmit point (TP), transmit-receive point (TRP), an enhanced base station (eNodeB or eNB), a 5G base station (gNB), a macrocell, a femtocell, a WiFi access point (AP), or other wirelessly enabled devices. Base stations may provide wireless access in accordance with one or more wireless communication protocols, e.g., 5G 3GPP new radio interface/access (NR), long term evolution (LTE), LTE advanced (LTE-A), high speed packet access (HSPA), Wi-Fi 802.11a/b/g/n/ac, etc. For the sake of convenience, the terms “BS” and “TRP” are used interchangeably in this patent document to refer to network infrastructure components that provide wireless access to remote terminals. Also, depending on the network type, the term “user equipment” or “UE” can refer to any component such as “mobile station,” “subscriber station,” “remote terminal,” “wireless terminal,” “receive point,” or “user device.” For the sake of convenience, the terms “user equipment” and “UE” are used in this patent document to refer to remote wireless equipment that wirelessly accesses a BS, whether the UE is a mobile device (such as a mobile telephone or smartphone) or is normally considered a stationary device (such as a desktop computer or vending machine).

(36) Dotted lines show the approximate extents of the coverage areas **120** and **125**, which are shown as approximately circular for the purposes of illustration and explanation only. It should be clearly understood that the coverage areas associated with gNBs, such as the coverage areas **120** and **125**, may have other shapes, including irregular shapes, depending upon the configuration of the gNBs and variations in the radio environment associated with natural and man-made obstructions.

(37) As described in more detail below, one or more of the UEs **111-116** include circuitry, programing, or a combination thereof, for

receiving information for a maximum permissible emission (MPE) report; determining the MPE report including Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$; including the MPE report in a PHR report; and transmitting, via MAC CE, the PHR report including the MPE report. One or more of the gNBs **101-103** includes circuitry, programing, or a combination thereof, for generating information for a MPE report; transmitting the information; and receiving, via a MAC CE, a PHR report, the PHR report including the MPE report, wherein the MPE report includes Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$.

(38) Although FIG. 1 illustrates one example of a wireless network, various changes may be made to FIG. 1. For example, the wireless network could include any number of gNBs and any number of UEs in any suitable arrangement. Also, the gNB **101** could communicate directly with any number of UEs and provide those UEs with wireless broadband access to the network **130**. Similarly, each gNB **102-103** could communicate directly with the network **130** and provide UEs with direct wireless broadband access to the network **130**. Further, the gNBs **101**, **102**, and/or **103** could provide access to other or additional external networks, such as external telephone networks or other types of data networks.

(39) FIG. 2 illustrates an example gNB **102** according to embodiments of the present disclosure. The embodiment of the gNB **102** illustrated in FIG. 2 is for illustration only, and the gNBs **101** and **103** of FIG. 1 could have the same or similar configuration. However, gNBs come in a wide variety of configurations, and FIG. 2 does not limit the scope of this disclosure to any particular implementation of a gNB.

(40) As shown in FIG. 2, the gNB **102** includes multiple antennas **205a-205n**, multiple RF transceivers **210a-210n**, transmit (TX) processing circuitry **215**, and receive (RX) processing circuitry **220**. The gNB **102** also includes a controller/processor **225**, a memory **230**, and a backhaul or network interface **235**.

(41) The RF transceivers **210a-210n** receive, from the antennas **205a-205n**, incoming RF signals, such as signals transmitted by UEs in the network **100**. The RF transceivers **210a-210n** down-convert the incoming RF signals to generate IF or baseband signals. The IF or baseband signals are sent to the RX processing circuitry **220**, which generates processed baseband signals by filtering, decoding, and/or digitizing the baseband or IF signals. The RX processing circuitry **220** transmits the processed baseband signals to the controller/processor **225** for further processing.

(42) The TX processing circuitry **215** receives analog or digital data (such as voice data, web data, e-mail, or interactive video game data) from the controller/processor **225**. The TX processing circuitry **215** encodes, multiplexes, and/or digitizes the outgoing baseband data to generate processed baseband or IF signals. The RF transceivers **210a-210n** receive the outgoing processed baseband or IF signals from the TX processing circuitry **215** and up-converts the baseband or IF signals to RF signals that are transmitted via the antennas **205a-205n**.

(43) The controller/processor **225** can include one or more processors or other processing devices that control the overall operation of the gNB **102**. For example, the controller/processor **225** could control the reception of UL channel signals and the transmission of DL channel signals by the RF transceivers **210a-210n**, the RX processing circuitry **220**, and the TX processing circuitry **215** in accordance with well-known principles. The controller/processor **225** could support additional functions as well, such as more advanced wireless communication functions.

(44) For instance, the controller/processor **225** could support beam forming or directional routing operations in which outgoing signals from multiple antennas **205a-205n** are weighted differently to effectively steer the outgoing signals in a desired direction. Any of a wide variety of other functions could be supported in the gNB **102** by the controller/processor **225**.

(45) The controller/processor **225** is also capable of executing programs and other processes resident in the memory **230**, such as an OS. The controller/processor **225** can move data into or out of the memory **230** as required by an executing process.

(46) The controller/processor **225** is also coupled to the backhaul or network interface **235**. The backhaul or network interface **235** allows the gNB **102** to communicate with other devices or systems over a backhaul connection or over a network. The interface **235** could support communications over any suitable wired or wireless connection(s). For example, when the gNB **102** is implemented as part of a cellular communication system (such as one supporting 5G, LTE, or LTE-A), the interface **235** could allow the gNB **102** to communicate with other gNBs over a wired or wireless backhaul connection. When the gNB **102** is implemented as an access point, the interface **235** could allow the gNB **102** to communicate over a wired or wireless local area network or over a wired or wireless connection to a larger network (such as the Internet). The interface **235** includes any suitable structure supporting communications over a wired or wireless connection, such as an Ethernet or RF transceiver.

(47) The memory **230** is coupled to the controller/processor **225**. Part of the memory **230** could include a RAM, and another part of the memory **230** could include a Flash memory or other ROM.

(48) Although FIG. 2 illustrates one example of gNB **102**, various changes may be made to FIG. 2. For example, the gNB **102** could include any number of each component shown in FIG. 2. As a particular example, an access point could include a number of interfaces **235**, and the controller/processor **225** could support routing functions to route data between different network addresses. As another particular example, while shown as including a single instance of TX processing circuitry **215** and a single instance of RX processing circuitry **220**, the gNB **102** could include multiple instances of each (such as one per RF transceiver). Also, various components in FIG. 2 could be combined, further subdivided, or omitted and additional components could be added according to particular needs.

(49) FIG. 3 illustrates an example UE **116** according to embodiments of the present disclosure. The embodiment of the UE **116** illustrated in FIG. 3 is for illustration only, and the UEs **111-115** of FIG. 1 could have the same or similar configuration. However, UEs come in a wide variety of configurations, and FIG. 3 does not limit the scope of this disclosure to any particular implementation of a UE.

(50) As shown in FIG. 3, the UE **116** includes an antenna **305**, a radio frequency (RF) transceiver **310**, TX processing circuitry **315**, a microphone **320**, and receive (RX) processing circuitry **325**. The UE **116** also includes a speaker **330**, a processor **340**, an input/output (I/O) interface (IF) **345**, a touchscreen **350**, a display **355**, and a memory **360**. The memory **360** includes an operating system (OS) **361** and one or more applications **362**.

(51) The RF transceiver **310** receives, from the antenna **305**, an incoming RF signal transmitted by a gNB of the network **100**. The RF transceiver **310** down-converts the incoming RF signal to generate an intermediate frequency (IF) or baseband signal. The IF or baseband signal is sent to the RX processing circuitry **325**, which generates a processed baseband signal by filtering, decoding, and/or digitizing the baseband or IF signal. The RX processing circuitry **325** transmits the processed baseband signal to the speaker **330** (such as for voice data) or to the processor **340** for further processing (such as for web browsing data).

(52) The TX processing circuitry **315** receives analog or digital voice data from the microphone **320** or other outgoing baseband data (such as web data, e-mail, or interactive video game data) from the processor **340**. The TX processing circuitry **315** encodes, multiplexes, and/or digitizes the outgoing baseband data to generate a processed baseband or IF signal. The RF transceiver **310** receives the outgoing processed baseband or IF signal from the TX processing circuitry **315** and up-converts the baseband or IF signal to an RF signal that is transmitted via the antenna **305**.

(53) The processor **340** can include one or more processors or other processing devices and execute the OS **361** stored in the memory **360** in order to control the overall operation of the UE **116**. For example, the processor **340** could control the reception of DL channel signals and the transmission of UL channel signals by the RF transceiver **310**, the RX processing circuitry **325**, and the TX processing circuitry **315** in accordance with well-known principles. In some embodiments, the processor **340** includes at least one microprocessor or microcontroller.

(54) The processor **340** is also capable of executing other processes and programs resident in the memory **360**, such as processes for receiving information for a maximum permissible emission (MPE) report; determining the MPE report including Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$; including the MPE report in a PHR report; and transmitting, via MAC CE, the PHR report including the MPE report. The processor **340** can move data into or out of the memory **360** as required by an executing process. In some embodiments, the processor **340** is configured to execute the applications **362** based on the OS **361** or in response to signals received from gNBs or an operator. The processor **340** is also coupled to the I/O interface **345**, which provides the UE **116** with the ability to connect to other devices, such as laptop computers and handheld computers. The I/O interface **345** is the communication path between these accessories and the processor **340**.

(55) The processor **340** is also coupled to the touchscreen **350** and the display **355**. The operator of the UE **116** can use the touchscreen **350** to enter data into the UE **116**. The display **355** may be a liquid crystal display, light emitting diode display, or other display capable of rendering text and/or at least limited graphics, such as from web sites.

(56) The memory **360** is coupled to the processor **340**. Part of the memory **360** could include a random-access memory (RAM), and another part of the memory **360** could include a Flash memory or other read-only memory (ROM).

(57) Although FIG. 3 illustrates one example of UE **116**, various changes may be made to FIG. 3. For example, various components in FIG. 3 could be combined, further subdivided, or omitted and additional components could be added according to particular needs. As a particular example, the processor **340** could be divided into multiple processors, such as one or more central processing units (CPUs) and one or more graphics processing units (GPUs). Also, while FIG. 3 illustrates the UE **116** configured as a mobile telephone or smartphone, UEs could be configured to operate as other types of mobile or stationary devices.

(58) FIG. 4A is a high-level diagram of transmit path circuitry. For example, the transmit path circuitry may be used for an orthogonal frequency division multiple access (OFDMA) communication. FIG. 4B is a high-level diagram of receive path circuitry. For example, the receive path circuitry may be used for an orthogonal frequency division multiple access (OFDMA) communication. In FIGS. 4A and 4B, for downlink communication, the transmit path circuitry may be implemented in a base station (gNB) **102** or a relay station, and the receive path circuitry may be implemented in a user equipment (e.g., user equipment **116** of FIG. 1). In other examples, for uplink communication, the receive path circuitry **450** may be implemented in a base station (e.g., gNB **102** of FIG. 1) or a relay station, and the transmit path circuitry may be implemented in a user equipment (e.g., user equipment **116** of FIG. 1).

(59) Transmit path circuitry comprises channel coding and modulation block **405**, serial-to-parallel (S-to-P) block **410**, Size N Inverse Fast Fourier Transform (IFFT) block **415**, parallel-to-serial (P-to-S) block **420**, add cyclic prefix block **425**, and up-converter (UC) **430**. Receive path circuitry **450** comprises down-converter (DC) **455**, remove cyclic prefix block **460**, serial-to-parallel (S-to-P) block **465**, Size N Fast Fourier Transform (FFT) block **470**, parallel-to-serial (P-to-S) block **475**, and channel decoding and demodulation block **480**.

(60) At least some of the components in FIGS. 4A **400** and 4B **450** may be implemented in software, while other components may be implemented by configurable hardware or a mixture of software and configurable hardware. In particular, it is noted that the FFT blocks and the IFFT blocks described in this disclosure document may be implemented as configurable software algorithms, where the value of Size N may be modified according to the implementation.

(61) Furthermore, although this disclosure is directed to an embodiment that implements the Fast Fourier Transform and the Inverse Fast Fourier Transform, this is by way of illustration only and may not be construed to limit the scope of the disclosure. It may be appreciated that in an alternate embodiment of the present disclosure, the Fast Fourier Transform functions and the Inverse Fast Fourier Transform functions may easily be replaced by discrete Fourier transform (DFT) functions and inverse discrete Fourier transform (IDFT) functions, respectively. It may be appreciated that for DFT and IDFT functions, the value of the N variable may be any integer number (i.e., 1, 4, 3, 4, etc.), while for FFT and IFFT functions, the value of the N variable may be any integer number that is a power of two (i.e., 1, 2, 4, 8, 16, etc.).

(62) In transmit path circuitry **400**, channel coding and modulation block **405** receives a set of information bits, applies coding (e.g., LDPC coding) and modulates (e.g., quadrature phase shift keying (QPSK) or quadrature amplitude modulation (QAM)) the input bits to produce a sequence of frequency-domain modulation symbols. Serial-to-parallel block **410** converts (i.e., de-multiplexes) the serial modulated symbols to parallel data to produce N parallel symbol streams where N is the IFFT/FFT size used in BS **102** and UE **116**. Size N IFFT block **415** then performs an IFFT operation on the N parallel symbol streams to produce time-domain output signals. Parallel-to-serial block **420** converts (i.e., multiplexes) the parallel time-domain output symbols from Size N IFFT block **415** to produce a serial time-domain signal. Add cyclic prefix block **425** then inserts a cyclic prefix to the time-domain signal. Finally, up-converter **430** modulates (i.e., up-converts) the output of add cyclic prefix block **425** to RF frequency for transmission via a wireless channel. The signal may also be filtered at baseband before conversion to RF frequency.

(63) The transmitted RF signal arrives at the UE **116** after passing through the wireless channel, and reverse operations to those at gNB **102** are performed. Down-converter **455** down-converts the received signal to baseband frequency and removes cyclic prefix block **460** and removes the cyclic prefix to produce the serial time-domain baseband signal. Serial-to-parallel block **465** converts the time-domain baseband signal to parallel time-domain signals. Size N FFT block **470** then performs an FFT algorithm to produce N parallel frequency-domain signals. Parallel-to-serial block **475** converts the parallel frequency-domain signals to a sequence of modulated data symbols. Channel decoding and demodulation block **480** demodulates and then decodes the modulated symbols to recover the original input data stream.

(64) Each of gNBs **101-103** may implement a transmit path that is analogous to transmitting in the downlink to user equipment **111-116** and may implement a receive path that is analogous to receiving in the uplink from user equipment **111-116**. Similarly, each one of user equipment **111-116** may implement a transmit path corresponding to the architecture for transmitting in the uplink to gNBs **101-103** and may implement a receive path corresponding to the architecture for receiving in the downlink from gNBs **101-103**.

(65) 5G communication system use cases have been identified and described. Those use cases can be roughly categorized into three different groups. In one example, enhanced mobile broadband (eMBB) is determined to do with high bits/sec requirement, with less stringent latency and reliability requirements. In another example, ultra-reliable and low latency (URLL) is determined with less stringent bits/sec requirement. In yet another example, massive machine type communication (mMTC) is determined that a number of devices can be as many as 100,000 to 1 million per km², but the reliability/throughput/latency requirement could be less stringent. This scenario may

also involve power efficiency requirement as well, in that the battery consumption may be minimized as possible.

(66) A communication system includes a downlink (DL) that conveys signals from transmission points such as base stations (BSs) or NodeBs to user equipments (UEs) and an Uplink (UL) that conveys signals from UEs to reception points such as NodeBs. A UE, also commonly referred to as a terminal or a mobile station, may be fixed or mobile and may be a cellular phone, a personal computer device, or an automated device. An eNodeB, which is generally a fixed station, may also be referred to as an access point or other equivalent terminology. For LTE systems, a NodeB is often referred to as an eNodeB.

(67) In a communication system, such as LTE system, DL signals can include data signals conveying information content, control signals conveying DL control information (DCI), and reference signals (RS) that are also known as pilot signals. An eNodeB transmits data information through a physical DL shared channel (PDSCH). An eNodeB transmits DCI through a physical DL control channel (PDCCH) or an Enhanced PDCCH (EPDCCH).

(68) An eNodeB transmits acknowledgement information in response to data transport block (TB) transmission from a UE in a physical hybrid ARQ indicator channel (PHICH). An eNodeB transmits one or more of multiple types of RS including a UE-common RS (CRS), a channel state information RS (CSI-RS), or a demodulation RS (DMRS). A CRS is transmitted over a DL system bandwidth (BW) and can be used by UEs to obtain a channel estimate to demodulate data or control information or to perform measurements. To reduce CRS overhead, an eNodeB may transmit a CSI-RS with a smaller density in the time and/or frequency domain than a CRS. DMRS can be transmitted only in the BW of a respective PDSCH or EPDCCH and a UE can use the DMRS to demodulate data or control information in a PDSCH or an EPDCCH, respectively. A transmission time interval for DL channels is referred to as a subframe and can have, for example, duration of 1 millisecond.

(69) DL signals also include transmission of a logical channel that carries system control information. A BCCH is mapped to either a transport channel referred to as a broadcast channel (BCH) when the DL signals convey a master information block (MIB) or to a DL shared channel (DL-SCH) when the DL signals convey a System Information Block (SIB). Most system information is included in different SIBs that are transmitted using DL-SCH. A presence of system information on a DL-SCH in a subframe can be indicated by a transmission of a corresponding PDCCH conveying a codeword with a cyclic redundancy check (CRC) scrambled with system information RNTI (SI-RNTI). Alternatively, scheduling information for a SIB transmission can be provided in an earlier SIB and scheduling information for the first SIB (SIB-1) can be provided by the MIB.

(70) DL resource allocation is performed in a unit of subframe and a group of physical resource blocks (PRBs). A transmission BW includes frequency resource units referred to as resource blocks (RBs). Each RB includes $N_{\text{sub.sc.sup.RB}}$ sub-carriers, or resource elements (REs), such as 12 REs. A unit of one RB over one subframe is referred to as a PRB. $M_{\text{sub.PDSCH}}$ RBs for a total of $M_{\text{sub.sc.sup.PDSCH}} = M_{\text{sub.PDSCH}} \cdot N_{\text{sub.sc.sup.RB}}$ REs for the PDSCH transmission BW.

(71) UL signals can include data signals conveying data information, control signals conveying UL control information (UCI), and UL RS. UL RS includes DMRS and Sounding RS (SRS). A UE transmits DMRS only in a BW of a respective PUSCH or PUCCH. An eNodeB can use a DMRS to demodulate data signals or UCI signals. A UE transmits SRS to provide an eNodeB with an UL CSI. A UE transmits data information or UCI through a respective physical UL shared channel (PUSCH) or a Physical UL control channel (PUCCH). If a UE needs to transmit data information and UCI in a same UL subframe, the UE may multiplex both in a PUSCH. UCI includes Hybrid Automatic Repeat request acknowledgement (HARQ-ACK) information, indicating correct (ACK) or incorrect (NACK) detection for a data TB in a PDSCH or absence of a PDCCH detection (DTX), scheduling request (SR) indicating whether a UE has data in the UE's buffer, rank indicator (RI), and channel state information (CSI) enabling an eNodeB to perform link adaptation for PDSCH transmissions to a UE. HARQ-ACK information is also transmitted by a UE in response to a detection of a PDCCH/EPDCCH indicating a release of semi-persistently scheduled PDSCH.

(72) An UL subframe includes two slots. Each slot includes $N_{\text{sub.sym.sup.UL}}$ UL symbols for transmitting data information, UCI, DMRS, or SRS. A frequency resource unit of an UL system BW is an RB. A UE is allocated $N_{\text{sub.RB}}$ RBs for a total of $N_{\text{sub.RB}} \cdot N_{\text{sub.sc.sup.RB}}$ REs for a transmission BW. For a PUCCH, $N_{\text{sub.RB}} = 1$. A last subframe symbol can be used to multiplex SRS transmissions from one or more UEs. A number of subframe symbols that are available for data/UCI/DMRS transmission is $N_{\text{sub.symb}} = 2 \cdot N_{\text{sub.sym.sup.UL}} - N_{\text{sub.SRS}}$, where $N_{\text{sub.SRS}} = 1$ if a last subframe symbol is used to transmit SRS and $N_{\text{sub.SRS}} = 0$ otherwise.

(73) FIG. 5 illustrates a transmitter block diagram **500** for a PDSCH in a subframe according to embodiments of the present disclosure. The embodiment of the transmitter block diagram **500** illustrated in FIG. 5 is for illustration only. One or more of the components illustrated in FIG. 5 can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 5 does not limit the scope of this disclosure to any particular implementation of the transmitter block diagram **500**.

(74) As shown in FIG. 5, information bits **510** are encoded by encoder **520**, such as a turbo encoder, and modulated by modulator **530**, for example using quadrature phase shift keying (QPSK) modulation. A serial to parallel (S/P) converter **540** generates M modulation symbols that are subsequently provided to a mapper **550** to be mapped to REs selected by a transmission BW selection unit **555** for an assigned PDSCH transmission BW, unit **560** applies an Inverse fast Fourier transform (IFFT), the output is then serialized by a parallel to serial (P/S) converter **570** to create a time domain signal, filtering is applied by filter **580**, and a signal transmitted **590**. Additional functionalities, such as data scrambling, cyclic prefix insertion, time windowing, interleaving, and others are well known in the art and are not shown for brevity.

(75) FIG. 6 illustrates a receiver block diagram **600** for a PDSCH in a subframe according to embodiments of the present disclosure. The embodiment of the diagram **600** illustrated in FIG. 6 is for illustration only. One or more of the components illustrated in FIG. 6 can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 6 does not limit the scope of this disclosure to any particular implementation of the diagram **600**.

(76) As shown in FIG. 6, a received signal **610** is filtered by filter **620**, REs **630** for an assigned reception BW are selected by BW selector **635**, unit **640** applies a fast Fourier transform (FFT), and an output is serialized by a parallel-to-serial converter **650**. Subsequently, a demodulator **660** coherently demodulates data symbols by applying a channel estimate obtained from a DMRS or a CRS (not shown), and a decoder **670**, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits **680**. Additional functionalities such as time-windowing, cyclic prefix removal, de-scrambling, channel estimation, and de-interleaving are not shown for brevity.

(77) FIG. 7 illustrates a transmitter block diagram **700** for a PUSCH in a subframe according to embodiments of the present disclosure. The embodiment of the block diagram **700** illustrated in FIG. 7 is for illustration only. One or more of the components illustrated in FIG. 5

can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 7 does not limit the scope of this disclosure to any particular implementation of the block diagram 700.

(78) As shown in FIG. 7, information data bits 710 are encoded by encoder 720, such as a turbo encoder, and modulated by modulator 730. A discrete Fourier transform (DFT) unit 740 applies a DFT on the modulated data bits, REs 750 corresponding to an assigned PUSCH transmission BW are selected by transmission BW selection unit 755, unit 760 applies an IFFT and, after a cyclic prefix insertion (not shown), filtering is applied by filter 770 and a signal transmitted 780.

(79) FIG. 8 illustrates a receiver block diagram 800 for a PUSCH in a subframe according to embodiments of the present disclosure. The embodiment of the block diagram 800 illustrated in FIG. 8 is for illustration only. One or more of the components illustrated in FIG. 8 can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. FIG. 8 does not limit the scope of this disclosure to any particular implementation of the block diagram 800.

(80) As shown in FIG. 8, a received signal 810 is filtered by filter 820. Subsequently, after a cyclic prefix is removed (not shown), unit 830 applies an FFT, REs 840 corresponding to an assigned PUSCH reception BW are selected by a reception BW selector 845, unit 850 applies an inverse DFT (IDFT), a demodulator 860 coherently demodulates data symbols by applying a channel estimate obtained from a DMRS (not shown), a decoder 870, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits 880.

(81) In next generation cellular systems, various use cases are envisioned beyond the capabilities of LTE system. Termed 5G or the fifth-generation cellular system, a system capable of operating at sub-6 GHz and above-6 GHz (for example, in mmWave regime) becomes one of the requirements. In 3GPP TR 22.891, 74 5G use cases have been identified and described; those use cases can be roughly categorized into three different groups. A first group is termed “enhanced mobile broadband (eMBB),” targeted to high data rate services with less stringent latency and reliability requirements. A second group is termed “ultra-reliable and low latency (URLL)” targeted for applications with less stringent data rate requirements, but less tolerant to latency. A third group is termed “massive MTC (mMTC)” targeted for large number of low-power device connections such as 1 million per km.sup.2 with less stringent the reliability, data rate, and latency requirements.

(82) The 3GPP NR specification supports up to 32 CSI-RS antenna ports which enable a gNB to be equipped with a large number of antenna elements (such as 64 or 128). In this case, a plurality of antenna elements is mapped onto one CSI-RS port. For next generation cellular systems such as 5G, the maximum number of CSI-RS ports can either remain the same or increase.

(83) FIG. 9 illustrates an example antenna blocks or arrays 900 according to embodiments of the present disclosure. The embodiment of the antenna blocks or arrays 900 illustrated in FIG. 9 is for illustration only. FIG. 9 does not limit the scope of this disclosure to any particular implementation of the antenna blocks or arrays 900.

(84) For mmWave bands, although the number of antenna elements can be larger for a given form factor, the number of CSI-RS ports—which can correspond to the number of digitally precoded ports—tends to be limited due to hardware constraints (such as the feasibility to install a large number of ADCs/DACs at mmWave frequencies) as illustrated in FIG. 9. In this case, one CSI-RS port is mapped onto a large number of antenna elements which can be controlled by a bank of analog phase shifters 901. One CSI-RS port can then correspond to one sub-array which produces a narrow analog beam through analog beamforming 905. This analog beam can be configured to sweep across a wider range of angles (920) by varying the phase shifter bank across symbols or subframes. The number of sub-arrays (equal to the number of RF chains) is the same as the number of CSI-RS ports N.sub.CSI-PORT. A digital beamforming unit 910 performs a linear combination across N.sub.CSI-PORT analog beams to further increase precoding gain. While analog beams are wideband (hence not frequency-selective), digital precoding can be varied across frequency sub-bands or resource blocks.

(85) Since the above system utilizes multiple analog beams for transmission and reception (wherein one or a small number of analog beams are selected out of a large number, for instance, after a training duration—to be performed from time to time), the term “multi-beam operation” is used to refer to the overall system aspect. This includes, for the purpose of illustration, indicating the assigned DL or UL transmit (TX) beam (also termed “beam indication”), measuring at least one reference signal for calculating and performing beam reporting (also termed “beam measurement” and “beam reporting”, respectively), and receiving a DL or UL transmission via a selection of a corresponding receive (RX) beam.

(86) The above system is also applicable to higher frequency bands such as >52.6 GHz (also termed the FR4). In this case, the system can employ only analog beams. Due to the O2 absorption loss around 60 GHz frequency (~10 dB additional loss @100 m distance), larger number of and sharper analog beams (hence larger number of radiators in the array) will be needed to compensate for the additional path loss.

(87) In Rel.15 NR, multi-beam operation is designed primarily for single transmit-receive point (TRP) and single antenna panel. Therefore, the specification supports beam indication for one TX beam wherein a TX beam is associated with a reference RS. For DL beam indication and measurement, the reference RS can be NZP (non-zero power) CSI-RS and/or SSB (synchronization signal block, which includes primary synchronization signal, secondary synchronization signal, and PBCH). Here, DL beam indication is done via the transmission configuration indicator (TCI) field in DL-related DCI which includes an index to one (and only one) assigned reference RS. A set of hypotheses or the so-called TCI states is configured via higher-layer (RRC) signaling and, when applicable, a subset of those TCI states is selected/activated via MAC CE for the TCI field code points. For UL beam indication and measurement, the reference RS can be NZP CSI-RS, SSB, and/or SRS. Here, UL beam indication is done via the SRS resource indicator (SRI) field in UL-related DCI which is linked to one (and only one) reference RS. This linkage is configured via higher-layer signaling using the SpatialRelationInfo RRC parameter. Essentially, only one TX beam is indicated to the UE.

(88) In one example pertinent to Rel.15/16 NR, beam management was designed to share the same framework as CSI acquisition. This, however, compromises the performance of beam management especially for FR2. This is because beam management operates mainly with analog beams (characteristic of FR2) which paradigmatically differ from CSI acquisition (designed with FR1 in mind). Consequently, Rel.15/16 beam management becomes cumbersome and is unlikely able to keep up with more aggressive use cases which require large number of beams and fast beam switching (e.g., higher frequency bands, high mobility, and/or larger number of narrower analog beams). In addition, Rel.15/16 was designed to accommodate a number of unknown or rudimentary capabilities (e.g., UEs not capable of beam correspondence). To be flexible, it results in a number of options. This becomes burdensome to L1 control signaling and therefore a number of reconfigurations are performed via RRC signaling (higher-layer configuration). While this avoids L1 control overhead, it either results in high latency (if reconfiguration is performed sparsely) or imposes high usage of PDSCH (since RRC signaling consumes PDSCH resources).

(89) In one example pertinent to Rel.15/16 NR, when beam correspondence is utilized, UL beam selection can be performed via measuring

DL-RS (CSI-RS and/or SSB) and CRI reporting accompanied with corresponding beam metrics (e.g., RSRP, SINR). That is, based on the CRI/RSRP or CRI/SINR reporting from the UE, the network (NW) can assume that the UE performs UL transmission on PUSCH with the UL TX beam associated with one of the latest CRI reports (especially the one with the highest RSRP or SINR). Likewise, the UE can assume that the NW knows about this UE selection. Therefore, there is no need for a separate UL beam indication (e.g., via SRI field or UL-TCI field in the respective UL grant).

(90) In Rel.15/16 NR, when beam correspondence is not utilized, UL beam selection can be performed via the NW selecting the UL TX beam and indicating it to the UE via the UL grant (signaled via SRI field or UL-TCI field—essentially indicating the UL TCI state associated with the UL TX beam). This selection is enabled by measuring the SRS transmitted from the UE (configured by the NW).

(91) In either case, when an event that results in the UE having to select an (alternate) UL TX beam different from what the NW expects, some additional mechanisms are needed to ensure that (a) the UE has the alternate UL TX beam available when the UE detects such an event and the next UL TX beam indication can only in a later time slot, and (b) the NW is aware of the UE decision. A few examples of such an event are as follows. In one example, such as event can happen due to the so-called Maximum Permissible Exposure (MPE) regulation, especially in North America, that restricts the UE transmission power in certain directions. That is, to prevent any excessive electromagnetic wave exposure on delicate soft tissues (e.g., brain tissues), the UE is to avoid transmitting high energy signal along some directions (e.g., toward the head). Unfortunately, such directions may correspond to the “best” UL TX beams (e.g., associated with the CRI of the highest reported RSRP/SINR, or associated with the SRS resource yielding the best measured SINR at the NW). When the “best” UL TX beams are not used for UL transmission, some loss of UL throughput (especially coverage) will occur. In another example, such as event can happen due to hardware (HW) limitation at a UE equipped with multiple antenna panels, and in response to the event, the UE needs to select/switch antenna panel for UL transmission. In yet another example, such an event can happen due to potential beam failure, and to avoid beam failure, the UE needs to select/switch antenna panel for UL transmission. In yet another example, such an event can happen due to sudden change in channel conditions (e.g., due to high speed, antenna/panel blockage etc.) which may result in beam failure, and the UE needs to switch/change TX beam in order to continue UL transmission without interruptions/failures or having to wait for the next UL TX beam update/indication.

(92) Therefore, there is a need for efficient designs for enabling UL TX beam selection in order to avoid outage (or beam failure), loss in UL throughput, loss in UL coverage, and issues related to HW, that may happen due to the events mentioned above. In this disclosure, several example embodiments are proposed for such designs.

(93) This disclosure covers several components which can be used in conjunction or in combination with one another or can operate as standalone schemes.

(94) In the present disclosure, the term “activation” describes an operation wherein a UE receives and decodes a signal from the network (or gNB) that signifies a starting point in time. The starting point can be a present or a future slot/subframe or symbol—the exact location either implicitly or explicitly indicated, or otherwise fixed or higher-layer configured. Upon successfully decoding the signal, the UE responds accordingly. The term “deactivation” describes an operation wherein a UE receives and decodes a signal from the network (or gNB) that signifies a stopping point in time. The stopping point can be a present or a future slot/subframe or symbol—the exact location either implicitly or explicitly indicated, or otherwise fixed or higher-layer configured. Upon successfully decoding the signal, the UE responds accordingly.

(95) Terminology such as TCI, TCI states, SpatialRelationInfo, target RS, reference RS, and other terms is used for illustrative purposes and therefore not normative. Other terms that refer to the same functions can also be used.

(96) A “reference RS” corresponds to a set of characteristics of DL or UL TX beam, such as direction, precoding/beamforming, number of ports, etc. For instance, as the UE receives a reference RS index/ID in a DL assigned represented by a TCI state, the UE applies the known characteristics of the reference RS to the assigned DL transmission. The reference RS can be received and measured by the UE (in this case, the reference RS is a downlink signal such as NZP CSI-RS and/or SSB) with the result of the measurement used for calculating a beam report (in Rel.15 NR, at least one L1-RSRP accompanied by at least one CRI). As the NW/gNB receives the beam report, the NW can be better equipped with information to assign a particular DL TX beam to the UE. Optionally, the reference RS can be transmitted by the UE (in this case, the reference RS is an uplink signal such as SRS). As the NW/gNB receives the reference RS, the NW/gNB can measure and calculate the needed information to assign a particular DL TX beam to the UE. This option is applicable when DL-UL beam pair correspondence holds.

(97) The reference RS can be dynamically triggered by the NW/gNB (e.g., via DCI in case of aperiodic RS), preconfigured with a certain time-domain behavior (such as periodicity and offset, in case of periodic RS), or a combination of such pre-configuration and activation/deactivation (in case of semi-persistent RS).

(98) There are two types of frequency range (FR) defined in 3GPP NR specifications. The sub-6 GHz range is called frequency range 1 (FR1) and millimeter wave range is called frequency range 2 (FR2). An example of the frequency range for FR1 and FR2 is shown below.

(99) TABLE-US-00001 Frequency range designation Corresponding frequency range FR1 450 MHz-6000 MHz FR2 24250 MHz-52600 MHz

(100) The following embodiment is an example of DL multi-beam operation that utilizes DL beam indication after the network (NW) receives some transmission from the UE. In the first example embodiment, aperiodic CSI-RS is transmitted by the NW and measured by the UE. Although aperiodic RS is used in these two examples, periodic or semi-persistent RS can also be used.

(101) For mmWave (or FR2) or higher frequency bands (such as >52.6 GHz or FR4) where multi-beam operation is especially relevant, transmission-reception process includes the receiver to select a receive (RX) beam for a given TX beam. For UL multi-beam operation, the gNB selects an UL RX beam for every UL TX beam (which corresponds to a reference RS). Therefore, when UL RS (such as SRS and/or DMRS) is used as reference RS, the NW/gNB triggers or configures the UE to transmit the UL RS (which is associated with a selection of UL TX beam). The gNB, upon receiving and measuring the UL RS, selects an UL RX beam. As a result, a TX-RX beam pair is derived. The NW/gNB can perform this operation for all the configured reference RSs (either per reference RS or “beam sweeping”) and determine all the TX-RX beam pairs associated with all the reference RSs configured to the UE. On the other hand, when DL RS (such as CSI-RS and/or SSB) is used as reference RS (pertinent when DL-UL beam correspondence or reciprocity holds), the NW/gNB transmit the RS to the UE (for UL and by reciprocity, this corresponds to an UL RX beam). In response, the UE measures the reference RS (and in the process selects an UL TX beam) and reports the beam metric associated with the quality of the reference RS. In this case, the UE determines the TX-RX beam pair for every configured (DL) reference RS. Therefore, although this knowledge is unavailable to the NW/gNB, the UE—upon receiving a reference RS (hence UL RX beam) indication from the NW/gNB—can select the UL TX beam from the knowledge on all the TX-RX beam pairs.

(102) In the present disclosure, the term “Resource Indicator”, also abbreviated as REI, is used to refer to an indicator of RS resource used

for signal/channel and/or interference measurement purposes and hence can be substituted with any other term that refers to the same function. Examples of REI include the aforementioned CSI-RS resource indicator (CRI) and SSB resource indicator (SSB-RI). Any other RS can also be used for signal/channel and/or interference measurement such as DMRS.

(103) FIG. 10 illustrates an uplink multi-beam operation 1000 according to embodiments of the present disclosure. The embodiment of the uplink multi-beam operation 1000 illustrated in FIG. 10 is for illustration only. FIG. 10 does not limit the scope of this disclosure to any particular implementation of the uplink multi-beam operation 1000.

(104) As illustrated in FIG. 10, an UL multi-beam operation 1000 starts with the gNB/NW signaling to a UE an aperiodic CSI-RS (AP-CSI-RS) trigger or indication (step 1001). This trigger or indication can be included in a DCI (either UL-related or DL-related, either separately or jointly signaled with an aperiodic CSI request/trigger) and indicate transmission of AP-CSI-RS in a same (zero-time offset) or later slot/sub-frame (>0 time offset). Upon receiving the AP-CSI-RS transmitted by the gNB/NW (step 1002), the UE measures the AP-CSI-RS and, in turn, calculates and reports a “beam metric” (indicating quality of a particular TX beam hypothesis) (step 1003). Examples of such beam reporting are CSI-RS resource indicator (CRI) or SSB resource indicator (SSB-RI) coupled with its associated L1-RSRP/L1-RSRQ/L1-SINR/CQI. Upon receiving the beam report from the UE, the NW can use the beam report to select an UL TX beam for the UE and indicate the UL TX beam selection (step 1004) using the SRI field in the UL-related DCI (that carries the UL grant, such as DCI format 0_1 in NR). The SRI corresponds to a “target” SRS resource that is linked to a reference RS (in this case, an AP-CSI-RS) via SpatialRelationInfo configuration. Upon successfully decoding the UL-related DCI with the SRI, the UE performs UL transmission (such as data transmission on PUSCH) with the UL TX beam associated with the SRI (step 1005).

(105) FIG. 11 illustrates an uplink multi-beam operation 1100 according to embodiments of the present disclosure. The embodiment of the uplink multi-beam operation 1100 illustrated in FIG. 11 is for illustration only. FIG. 11 does not limit the scope of this disclosure to any particular implementation of the uplink multi-beam operation 1100.

(106) As illustrated in FIG. 11, an UL multi-beam operation 1100 starts with the gNB/NW signaling to a UE an aperiodic SRS (AP-SRS) trigger or request (step 1101). This trigger can be included in a DCI (either UL-related or DL-related). Upon receiving and decoding the AP-SRS trigger (step 1102), the UE transmits AP-SRS to the gNB/NW (step 1103) so that the NW (or gNB) can measure the UL propagation channel and select an UL TX beam for the UE. The gNB/NW can then indicate the UL TX beam selection (step 1104) using the SRI field in the UL-related DCI (that carries the UL grant, such as DCI format 0_1 in NR). The SRI corresponds to a “target” SRS resource that is linked to a reference RS (in this case, an AP-SRS) via SpatialRelationInfo configuration. Upon successfully decoding the UL-related DCI with the SRI, the UE performs UL transmission (such as data transmission on PUSCH) with the UL TX beam associated with the SRI (step 1105).

(107) FIG. 12 illustrates a downlink multi-beam operation 1200 according to embodiments of the present disclosure. The embodiment of the downlink multi-beam operation 1200 illustrated in FIG. 12 is for illustration only. FIG. 12 does not limit the scope of this disclosure to any particular implementation of the downlink multi-beam operation 1200.

(108) As illustrated in FIG. 12, wherein a UE is configured for measuring/receiving aperiodic CSI-RS (AP-CSI-RS) and reporting aperiodic CSI (AP CSI), a DL multi-beam operation 800 starts with the gNB/NW signaling to a UE an aperiodic CSI-RS (AP-CSI-RS) trigger or indication (step 1201). This trigger or indication can be included in a DCI (either UL-related or DL-related, either separately or jointly signaled with an aperiodic CSI request/trigger) and indicate transmission of AP-CSI-RS in a same (zero-time offset) or later slot/sub-frame (>0 time offset). Upon receiving the AP-CSI-RS transmitted by the gNB/NW (step 1202), the UE measures the AP-CSI-RS and, in turn, calculates and reports a “beam metric” (included in the CSI, indicating quality of a particular TX beam hypothesis) (step 1203). Examples of such beam reporting (supported in Rel.15/16 NR) are CSI-RS resource indicator (CRI) or SSB resource indicator (SSB-RI) coupled with its associated L1-RSRP and/or L1-SINR. Upon receiving the beam report from the UE, the NW/gNB can use the beam report to select a DL TX beam for the UE and indicate the DL TX beam selection (step 1204) using the TCI field in the DL-related DCI (that carries the DL assignment, such as DCI format 1_1 in NR). The TCI state corresponds to a reference RS (in this case, an AP-CSI-RS) defined/configured via the TCI state definition (higher-layer/RRC configured, from which a subset is activated via MAC CE for the DCI-based selection). Upon successfully decoding the DL-related DCI with the TCI field, the UE performs DL reception (such as data transmission on PDSCH) with the DL TX beam associated with the TCI field (step 1205). In this example embodiment, only one DL TX beam is indicated to the UE.

(109) The SRI used in the embodiments illustrated in FIGS. 10 and 11 can also be replaced with UL-TCI wherein an UL-TCI field can be introduced in the pertinent UL-related DCI(s), either in place of or in addition to the SRI field in Rel.15/16.

(110) The aperiodic CSI-RS (along with the associated aperiodic reporting) in the embodiment illustrated in FIG. 1000 and the aperiodic SRS illustrated in the embodiment illustrated in FIG. 11 can be substituted with that of another time-domain behavior such as semi-persistent (SP) or periodic (P).

(111) In any of the embodiments or sub-embodiments or examples below, a flowchart is used for illustrative purposes. The present disclosure covers any possible variation of the flowchart as long as at least some of the components are included. Such components include the UL TX beam indication indicating multiple UL TX beams and the event-dependent UL TX beam switch from the indicated multiple UL TX beams.

(112) In the rest of the disclosure, the term “beam”, can be associated with a spatial transmission/reception of a resource signal (RS) from a “port”, “antenna port”, or “virtual antenna/port”. Likewise, the term “transmit (TX) beam”, can be associated with a spatial transmission of a resource signal (RS) or a channel from a “port”, “antenna port”, or “virtual antenna/port”; and the term “receive (RX) beam”, can be associated with a spatial reception of a resource signal (RS) or a channel from a “port”, “antenna port”, or “virtual antenna/port”. The spatial transmission/reception of a beam can be in a three-dimension (3D) space. In a beam-formed wireless system, the transmission and reception of wireless signal can be via multiple TX and multiple RX beams.

(113) The present disclosure includes the following components for enabling beam measurement and reporting selection procedures, wherein the beam measurement and reporting may include an additional information about a state of each beam reporting, and the state can be related to MPE event, multiple UE panels, whether the beam report is for DL only, UL only or both DL and UL etc.

(114) FIG. 13 illustrates a flow diagram 1300 of a UE configured to measure P.sub.1 DL measurement RS resources according to embodiments of the present disclosure. The embodiment of the flow diagram 1300 of a UE configured to measure P.sub.1 DL measurement RS resources illustrated in FIG. 13 is for illustration only. FIG. 13 does not limit the scope of this disclosure to any particular implementation of the flow diagram 1300 of a UE configured to measure P.sub.1 DL measurement RS resources.

(115) In one embodiment (I.1), as illustrated in FIG. 13, a UE is configured (by the NW/gNB) to measure (receive) P.sub.1 DL measurement RS resources (such as CSI-RS or SSB), where $P_{sub.1} \geq 1$. This configuration can be performed via higher-layer (RRC) signaling. Optionally, the NW/gNB can signal/update the (sub)set of DL measurement RS resources dynamically via L1 or L2 DL control

(PDCCH or MAC CE). These resources are used by the UE to perform beam measurement along different beams or spatial directions (represented by the beamforming/precoding operation performed at the NW/gNB transparent to the UE).

(116) The UE is further configured (by the NW/gNB) to report a beam report, wherein the beam report includes Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), where $Q.sub.1 \leq P.sub.1$. In one example, $Q.sub.1=1$. In one example, $Q.sub.1$ is configured via RRC and/or MAC CE. The beam metric can represent link quality associated with the DL channel (or UL channel since beam correspondence holds). Examples of beam metric include L1-RSRP, L1-SINR, CQI, or hypothetical BLER, or virtual PHR, or UL RSRP, or any other beam metric. The resource indicator indicates a DL measurement RS resource index from the $P.sub.1$ DL measurement RS resources. Examples of resource indicator include CRI (when DL measurement RS is CSI-RS) and SSB-RI (when DL measurement RS is SSB).

(117) In one example, when the UE is equipped with $X>1$ antenna panels, $Q.sub.1=X$, and one resource indicator (I) or pair of (I,J)=(resource indicator, beam metric) is reported for each antenna panel. In one example, when the UE is equipped with $X>1$ antenna panels, $Q.sub.1 \geq X$, and at least one resource indicator (I) or pair of (I,J)=(resource indicator, beam metric) is reported for each antenna panel. The set of $P.sub.1$ DL measurement RS resources can be partitioned into X subsets, one subset for each antenna panel. Or the $P.sub.1$ DL measurement RS resources is a super-set encompassing X sets, one set for each antenna panel. In one example, the information about X panels is not provided to the NW/gNB. In one example, the information about X panels is provided to the NW/gNB. For instance, the information about the panel ID can be included/reported implicitly, e.g., from the resource indicators (I). Or the information about the panel ID can be included/reported explicitly, e.g., by including/reporting the panel ID(s) in the beam report.

(118) In addition to Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), the beam report can also include a UE-recommendation for UL TX beam selection/indication (by the NW/gNB). The time-domain behavior of this beam measurement and/or beam reporting can be configured as aperiodic (AP), semi-persistent (SP), or periodic (P). The time-domain behavior of the beam measurement RS can be configured as aperiodic (AP), semi-persistent (SP), or periodic (P).

(119) The UE-recommendation can be reported jointly with the other components included in the beam report using a joint parameter (or indicator). For example, the UE-recommendation can be reported jointly with at least one of the resource indicators. Or the UE-recommendation can be reported jointly with at least one of the beam metrics. Or the UE-recommendation can be reported jointly with at least one of the pairs (resource indicator, beam metric).

(120) Or the UE-recommendation can be reported separately (independently) from the other components in the beam report using a separate parameter (or indicator). The reporting configuration can be joint (one configuration) or separate (two configurations), one for UE-recommendation and another for the other components in the beam report.

(121) Or the reporting of the UE-recommendation and the other components in the beam report are decoupled, i.e., one reporting for the UE-recommendation and another reporting for the other components in the beam report. The reporting configuration can be joint (one configuration) or separate (two configurations), one for UE-recommendation and another for the other components in the beam report.

(122) Note that the UE-recommendation may not be limited to any specific event (such as MPE), it is rather general and is hence applicable to any event of interests for the UE such as MPE mitigation, fast panel selection, fast beam switch, avoiding beam failure (as described above).

(123) In one sub-embodiment (I.1.1), the content of the UE-recommendation is determined (or configured) according to at least one of the following examples.

(124) In one example (I.1.1.1), the UE-recommendation comprises Q.sub.2 additional (beam) resource indicator(s). In one example, $Q.sub.2$ is fixed, for example, to $Q.sub.2=1$. In one example, $Q.sub.2$ is configured. In one example, $Q.sub.2$ is reported by the UE, for example, as part of the beam report (i.e., UE-recommendation). In one example, a UE can report a $Q.sub.2$ value of zero, and in another example, a UE can report a non-zero (positive) value of $Q.sub.2$. In another example, the UE can be configured with a maximum value (v) for $Q.sub.2$, and the UE can report any $Q.sub.2$ value such that $Q.sub.2 \in \{1, 2, \dots, v\}$ or $\{0, 1, 2, \dots, v\}$.

(125) In one example (I.1.1.2), the UE-recommendation comprises Q.sub.2 additional pairs of (I,J)=(resource indicator, beam metric). In one example, $Q.sub.2$ is fixed, for example, to $Q.sub.2=1$. In one example, $Q.sub.2$ is configured. In one example, $Q.sub.2$ is reported by the UE, for example, as part of the beam report (i.e., UE-recommendation). In one example, a UE can report a $Q.sub.2$ value of zero, and in another example, a UE can report a non-zero (positive) value of $Q.sub.2$. In another example, the UE can be configured with a maximum value (v) for $Q.sub.2$, and the UE can report any $Q.sub.2$ value such that $Q.sub.2 \in \{1, 2, \dots, v\}$ or $\{0, 1, 2, \dots, v\}$.

(126) In one example (I.1.1.3), the UE-recommendation comprises an ID, where the ID can be associated with a subset of DL measurement RS resources. In one example, the set of $P.sub.1$ DL measurement RS resources can be partitioned into X subsets, and the x-th subset includes $M.sub.x$ DL measurement RS resources such that $P.sub.1 = \sum_{x=1}^{X} M.sub.x$. The ID (e.g., subset ID) indicates one of the X subsets. In one example, the $P.sub.1$ DL measurement RS resources is a super-set encompassing X sets, and the x-th set includes $M.sub.x$ DL measurement RS resources such that $P.sub.1 = \sum_{x=1}^{X} M.sub.x$. The ID (e.g., set ID) indicates one of the X sets. In one example, the ID can be associated with an antenna panel of the UE (e.g., panel ID) that is equipped with multiple antenna panels.

(127) In one example (I.1.1.4), the UE-recommendation comprises an ID and Q.sub.2 additional (beam) resource indicator(s), where the ID is according to the description in example (I.1.3), and Q.sub.2 additional (beam) resource indicator(s) is (are) according to the description in example (I.1.1). In one example, the Q.sub.2 additional (beam) resource indicator(s) can be associated with (dependent or derived from or based on) the ID. In one example, the Q.sub.2 additional (beam) resource indicator(s) can be independent of the ID.

(128) In one example (I.1.1.5), the UE-recommendation comprises an ID and Q.sub.2 additional pairs of (I,J)=(resource indicator, beam metric), where the ID is according to the description in example (I.1.3), and Q.sub.2 additional pairs of (I,J)=(resource indicator, beam metric) according to the description in example (I.1.2).

(129) In one sub-embodiment (I.1.2), the reporting of the UE-recommendation can be determined (or configured) according to at least one of the following examples.

(130) In one example (I.1.2.1), the UE-recommendation can always be included the beam report (i.e., reported by the UE).

(131) In one example (I.1.2.2), the UE-recommendation can always be included the beam report (i.e., reported by the UE) when a condition is met, wherein, for example, the condition being that the UE is equipped with multiple antenna panels, or that the UE reports this (the UE-recommendation) as part of its UE capability, or that an event of interest can be detected by the UE.

(132) In one example (I.1.2.3), the reporting of the UE-recommendation can be configured (turned ON/OFF), for example, via RRC and/or MAC CE and/or DCI. When configured (turned ON), the UE-recommendation is always included in the beam report.

(133) In one example (I.1.2.4), the UE decides (has the freedom to decide) whether the UE-recommendation is included in the beam report, i.e., the UE may or may not include the UE-recommendation in the beam report. For instance, such a decision can be made based on the event of interest. In one example, the beam report (with or without the UE-recommendation) is reported via a one-part UCI (similar to WB

CSI reporting in Rel. 15 NR, the UE can append a fixed number of zeros in order to ensure that the payload remains the same regardless of whether the UE reports the UE-recommendation or not). In one example, the beam report (with or without the UE-recommendation) is reported via a two-part UCI, wherein part 1 UCI includes an information whether the UE reports the UE-recommendation or not. If the information indicates the UE-recommendation being reported, it is reported via part 2 UCI.

(134) In one example (I.1.2.5), the capability of reporting of the UE-recommendation is reported by the UE (e.g., via UE capability reporting or as part of an UL transmission). Depending on the reported UE capability, the NW/gNB can configure (or trigger) the reporting of the UE-recommendation via RRC and/or MAC CE (and/or DCI). The UE reports the UE-recommendation according to the configuration/triggering from the NW/gNB.

(135) Upon receiving the beam report, the NW/gNB configures/indicates the UL TX beam indication (which can be the same as DL TX beam indication since beam correspondence holds) to the UE for UL transmission, where the beam indication indicates (A) a message, or (B) $N \geq 1$ UL TX beam(s), or (C) both a message and $N \geq 1$ UL TX beam(s). The beam indication can be via DL-TCI or UL-TCI or J-TCI (joint TCI) or other functionally equivalent entity such as SpatialRelationInfo or SRI that is indicated via DCI and/or MAC CE and/or RRC. In one example $N=1$. In one example, $N=2$.

(136) In one sub-embodiment (I.1.3), when the beam report does not include the UE-recommendation, the beam indication is determined (or configured) according to at least one of the following examples.

(137) In one example (I.1.3.1), when $Q_{\text{sub}.1}=1$ resource indicator is included in the beam report, the beam indication can only indicate (A) a message, where the message corresponds to an ACK message indicating that the $N=1$ UL TX beam corresponds to the $Q_{\text{sub}.1}$ resource indicator included in the beam report. Some of the details of the ACK message can be according to the same described in embodiment (I.4).

(138) In one example (I.1.3.2), when $Q_{\text{sub}.1}=1$ resource indicator is included in the beam report, the beam indication can indicate (A) a message or (B) $N \geq 1$ UL TX beam(s), where for (A), the message corresponds to an ACK message indicating that the $N=1$ UL TX beam corresponds to the $Q_{\text{sub}.1}$ resource indicator included in the beam report, and for (B), the $N \geq 1$ UL TX beam(s) is (are) based on the $P_{\text{sub}.1}$ DL measurement RS resources measured by the UE. Some of the details of the ACK message can be according to the same described in embodiment (I.4).

(139) In one example (I.1.3.3), the beam indication can only indicate (B) $N \geq 1$ UL TX beam(s), where the $N \geq 1$ UL TX beam(s) is (are) based on the $P_{\text{sub}.1}$ DL measurement RS resources measured by the UE.

(140) In one sub-embodiment (I.1.4), when the beam report includes the UE-recommendation, the NW/gNB can acknowledge to the UE whether it follows the UE-recommendation or not, and accordingly, the beam indication is determined (or configured) according to at least one of the following examples. Some of the details of the ACK message can be according to the same described in embodiment (I.4).

(141) In one example (I.1.4.1), the NW/gNB follows the UE-recommendation and the beam indication comprises (A) an ACK message to the UE. The NW/gNB may not indicate (B) $N \geq 1$ UL TX beam(s) in this case. Or the NW/gNB may indicate (B) $N \geq 1$ UL TX beam(s). Or, whether the NW/gNB indicates (B) $N \geq 1$ UL TX beam(s) or not may depend on the UE-recommendation. For example, when the UE-recommendation includes $Q_{\text{sub}.2}=1$ additional (beam) resource indicator (cf. example (I.1.1.1) and (I.1.1.2)), the NW/gNB doesn't indicate (B).

(142) In one example (I.1.4.2), the NW/gNB does not follow the UE-recommendation, it either doesn't send the ACK message or sends a NACK/NULL message to the UE. The NW/gNB indicates the beam indication comprising (B) $N \geq 1$ UL TX beam(s). The beam indication can be based on the $Q_{\text{sub}.1}$ resource indicators included in the beam report. Or the beam indication can be based on the $Q_{\text{sub}.1}$ resource indicators included in the beam report, and if included in the UE-recommendation, also based on the $Q_{\text{sub}.2}$ resource indicators.

(143) In an example, the NW/gNB signals the ACK message included in the beam indication only when the UE-recommendation is provided (reported) by the UE.

(144) In one sub-embodiment (I.1.5), the signaling the ACK (or NACK/NULL) message to the UE is determined (or configured) according to at least one of the following examples. Some of the details of the ACK message can be according to the same described in embodiment (I.4).

(145) In one example (I.1.5.1), the signaling of the ACK message is via a dedicated (separate) parameter or field. The dedicated parameter or field can be indicated via a DCI, where the DCI can be a UL-DCI (scheduling UL grant), or DL-DCI (scheduling DL transmission), or a separate DCI for the beam (TCI state) indication (UL-TCI-DCI or DL-TCI-DCI or TCI-DCI). Alternatively, the dedicated parameter or field can be indicated via channel other than DCI (e.g., PDSCH or MAC CE). In one example, the dedicated field corresponds to a 1-bit indication, where a bit value 0 indicates the ACK message (and, optionally, 1 indicates the NACK/NULL message), or vice versa, i.e., 1 indicates the ACK message (and, optionally, 0 indicates the NACK/NULL message). In one example, the dedicated parameter can take one of a value v_0 indicating the ACK message (and, optionally a value v_1 indicating the NACK/NULL message).

(146) In one example (I.1.5.2), the signaling of the ACK message is joint with a parameter or field, which can be indicated via a DCI or a channel other than DCI, as described above. In one example, the field corresponds to a B-bit indication, where when the B-bit indication equals a fixed bit sequence $b_{\text{sub}.0}b_{\text{sub}.1} \dots b_{\text{sub}.B-1}$, it indicates the ACK message. In one example, the fixed bit sequence is all-zero, i.e., $b_{\text{sub}.0}b_{\text{sub}.1} \dots b_{\text{sub}.B-1}=00 \dots 0$. In one example, the fixed bit sequence is all-one, i.e., $b_{\text{sub}.0}b_{\text{sub}.1} \dots b_{\text{sub}.B-1}=11 \dots 1$. For example, when $B=3$, i.e., a 3-bit indication is used for the beam indication, the code point 000 (or 111) can be used to indicate the ACK message. The remaining bit sequence values can be used indicate other DL or UL related parameter (such as $N \geq 1$ UL TX beam indication when the gNB/NW does not signal the ACK message).

(147) In one example (I.1.5.3), the signaling of the ACK (or NACK/NULL) message is configured (turned ON/OFF) to the UE via RRC and/or MAC CE. When configured (turned ON), the signaling is performed according to at least one of examples I.1.5.1 and I.1.5.2. Some of the details of the ACK message can be according to the same described in embodiment (I.4).

(148) In one example, the field (code point) value or the parameter value for the ACK (or NULL/NACK) message is reserved and can't be used for other purpose regardless of whether the UE is configured to report the ACK (or NULL/NACK) message or not.

(149) In one example, when configured (turned ON), the field (code point) value or the parameter value for the ACK (or NULL/NACK) message is used for indicating the ACK (or NULL/NACK) message, otherwise (turned OFF), it is used for indicating other DL or UL related parameter (such as $N \geq 1$ UL TX beam indication when the gNB/NW does not signal the ACK message).

(150) In one sub-embodiment (I.1.6), a UE provides two sets of reports, either with a same report or separately in two different reports, wherein; A first report (sub-report) is a beam report including resource indicator(s) and possibly beam metrics, without reflecting the UE's recommendation. A second report (sub-report) is a beam report including resource indicator(s) and possibly beam metrics, reflecting the UE's recommendation.

(151) In one example, this behavior is configurable/triggered to the UE via RRC and/or MAC CE and/or DCI. When configured, the UE

provides two reports (sub-reports), else it provides one report (the first report). In one example, the configuration and/or triggering is UE-specific, in another example, the configuration and/or triggering is UE-group specific, and in another example, the configuration and/or triggering is cell-specific.

(152) In one sub-embodiment (I.1.7), a UE is configured to report the UE-recommendation, as described above (and elsewhere in this disclosure), wherein its reporting is subject to a restriction. At least one of the following examples or a combination of multiple of the following examples is used (or configured) as the restriction.

(153) In one example (I.1.7.1), the restriction is on the resource type. The resource type can be CSI-RS or SSB for the beam report (including Q1 resource indicators or Q1 pairs of resource indicator and beam metric). However, for the UE-recommendation, it can be according to at least one of the following examples. In one example, the resource type can be CSI-RS or SSB for the UE-recommendation (including Q2 resource indicators or Q2 pairs of resource indicator and beam metric). In one example, the resource type can be CSI-RS for the UE-recommendation (including Q2 resource indicators or Q2 pairs of resource indicator and beam metric). In one example, the resource type can be SSB for the UE-recommendation (including Q2 resource indicators or Q2 pairs of resource indicator and beam metric).

(154) In one example (I.1.7.2), the restriction is based on the value of Q.sub.1. At least one of the following examples can be used. In one example, the UE-recommendation can only be reported when Q.sub.1=1, i.e., the reporting of one resource indicator (e.g., CRI or SSBRI) or one pair of resource indicator and beam metric (e.g., CRI/SSB-RI+RSRP/SINR) is configured. The UE-recommendation can include CRI/SSB-RI (or CRI/SSBRI+RSRP/SINR) associated (or QCL-ed) with UL TX beams that should be used even if they do not correspond to the beams with the largest RSRP/SINR. In one example, the UE-recommendation can only be reported when Q.sub.1=2, i.e., the reporting of two resource indicators (e.g., CRIs or SSBRI) or two pairs of resource indicator and beam metric (e.g., CRI/SSB-RI+RSRP/SINR) is configured. The UE-recommendation can include CRI/SSB-RI (or CRI/SSBRI+RSRP/SINR) associated (or QCL-ed) with UL TX beams that should be used even if they do not correspond to the beams with the largest RSRP/SINR. In one example, the UE-recommendation can only be reported when Q.sub.1=1 or 2, i.e., the reporting of one or two resource indicators (e.g., CRI or SSBRI) or one or two pairs of resource indicator and beam metric (e.g., CRI/SSB-RI+RSRP/SINR) is configured. The UE-recommendation can include CRI/SSB-RI (or CRI/SSBRI+RSRP/SINR) associated (or QCL-ed) with UL TX beams that should be used even if they do not correspond to the beams with the largest RSRP/SINR.

(155) In one example (I.1.7.3), the beam reporting including Q.sub.1 beam reports remains the same regardless of whether the UE-recommendation is reported or not, since it can be used for DL and/or UL (e.g., for beam indication for DL reception and/or UL transmission). However, the UE-recommendation when reported is used only for UL (e.g., for beam indication for UL transmission).

(156) FIG. 14 illustrates a flow diagram 1400 of a UE configured to measure P.sub.1 DL measurement RS resources according to embodiments of the present disclosure. The embodiment of the flow diagram 1400 of a UE configured to measure P.sub.1 DL measurement RS resources illustrated in FIG. 14 is for illustration only. FIG. 14 does not limit the scope of this disclosure to any particular implementation of the flow diagram 1400 of a UE configured to measure P.sub.1 DL measurement RS resources.

(157) In one embodiment (I.2), as illustrated in FIG. 14, a UE is configured (by the NW/gNB) to measure (receive) P.sub.1 DL measurement RS resources (such as CSI-RS or SSB), where P.sub.1≥1, and the details about this measurement are as described in embodiment (I.1).

(158) The UE is further configured (by the NW/gNB) to report a beam report, wherein the beam report includes Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), where Q.sub.1≤P.sub.1, and resource indicator and beam metric are according to the examples described in embodiment (I.1). For example, the resource indicator can be SSBRI or CRI, and the beam metric can be L1-RSRP or L1-SINR, as defined in Rel. 15/16 NR specification.

(159) In addition to Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), the beam report can also include an information about the ‘state’ of the beam reporting (e.g., state=MPE condition). The time-domain behavior of this beam measurement and/or beam reporting can be configured as aperiodic (AP), semi-persistent (SP), or periodic (P). The time-domain behavior of the beam measurement RS can be configured as aperiodic (AP), semi-persistent (SP), or periodic (P). The information about the ‘state’ can be reported via a bitmap or bit-field or bit sequence. Or the information about the ‘state’ can be reported via a parameter or information element (IE).

(160) The ‘state’ can be reported jointly with the other components included in the beam report using a joint parameter (or indicator). For example, the ‘state’ can be reported jointly with at least one of the resource indicators. Or the ‘state’ can be reported jointly with at least one of the beam metrics. Or the ‘state’ can be reported jointly with at least one of the pairs (resource indicator, beam metric).

(161) Or the ‘state’ can be reported separately (independently) from the other components in the beam report using a separate parameter (or indicator). The reporting configuration can be joint (one configuration) or separate (two configurations), one for ‘state’(s) and another for the other components in the beam report.

(162) Or the reporting of the ‘state’ and the other components in the beam report are decoupled, i.e., one reporting for the ‘state’ and another reporting for the other components in the beam report. The reporting configuration can be joint (one configuration) or separate (two configurations), one for ‘state’(s) and another for the other components in the beam report.

(163) In one example, when the UE is equipped with X>1 antenna panels, Q.sub.1=X, and one resource indicator (I) or pair of (I,J)=(resource indicator, beam metric) is reported for each antenna panel. In one example, when the UE is equipped with X>1 antenna panels, Q.sub.1≥X, and at least one resource indicator (I) or pair of (I,J)=(resource indicator, beam metric) is reported for each antenna panel. The set of P.sub.1 DL measurement RS resources can be partitioned into X subsets, one subset for each antenna panel. Or the P.sub.1 DL measurement RS resources is a super-set encompassing X sets, one set for each antenna panel. In one example, the information about X panels are not provided to the NW/gNB. In one example, the information about X panels is provided to the NW/gNB. For instance, the information about the panel ID can be included/reported implicitly, e.g., from the resource indicators (I). Or, the information about the panel ID can be included/reported explicitly, e.g., by including/reporting the panel ID(s) in the beam report.

(164) For X>1 antenna panels at the UE, the ‘state’ can be reported independently for each panel. Or the ‘state’ can be reported for one of the X panels (e.g., the panel that has the worst MPE issue), and can also include a corresponding panel ID optionally. Or the ‘state’ can be reported for all panels that have MPE issue. If there is no MPE issue detected, the UE does not report the ‘state’, otherwise, the UE reports the ‘state’ for all panels having the MPE issue and can also report the corresponding panels ID(s) optionally.

(165) For X>1 subsets (or sets) of DL measurement RS resources, the ‘state’ can be reported independently for each subset (or set). Or the ‘state’ can be reported for one of the X subsets (or sets), e.g., the subset or the set that has the worst MPE issue and can also include a corresponding subset (or set) ID optionally. Or the ‘state’ can be reported for all subsets (or sets) that have MPE issue. If there is no MPE issue detected, the UE does not report the ‘state’, otherwise, the UE reports the ‘state’ for all subsets (or sets) having the MPE issue and

can also report the corresponding subset (or set) ID(s) optionally.

(166) The 'state' can be related to an event of interest (such as MPE), or panel status (or ID), or whether the beam report is for one of or both of DL reception and UL transmission. At least one of the following examples can be used for reporting the information about the state. In example A, the state is regarding the MPE event or MPE mitigation. For example, a 1-bit indication B can be used to indicate one of the two states: (i) MPE detected (or MPE mitigation), and (ii) MPE not detected (not for MPE mitigation). For example, B=1 indicates (i) and B=0 indicates (ii), or vice versa. Alternatively, a parameter p is used to indicate one of the two states: (i) and (ii), via value $p=v_0$ and $p=v_1$, respectively. In example B, the state is regarding a panel entity when the UE is equipped with multiple UE panels. For example, the state can correspond to an ID e.g., panel ID or set ID or resource ID associated with a panel. In example C, the state is regarding a combination of the MPE event and panel entity. For example, the state can indicate a pair (x,y), where x is either (i) or (ii) in example A, and y is an ID as described in example B. In example D, the state is regarding the recommended/preferred usage of the beam report for one of both of DL reception and UL transmission. At least one of the following examples is used/configured. In one example, the state indicates one of the two possible values $S=\{\text{DL reception, none}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{DL reception, UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{DL reception, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{UL transmission, both DL reception and UL transmission}\}$. In one example, the state indicates one of the three possible values $S=\{\text{DL reception, UL transmission, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, DL reception, UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, DL reception, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, UL transmission, both DL reception and UL transmission}\}$. In one example, the state indicates one of the three possible values $S=\{\text{none, DL reception, UL transmission, both DL reception and UL transmission}\}$. In example E, the state is regarding a combination of the MPE event and the recommended usage of the beam report for one of both of DL reception and UL transmission. For example, the state can indicate a pair (x,y), where x is either (i) or (ii) in example A, and y takes a value from S, as described in example D. In example F, the state is regarding a combination of the panel entity and the recommended usage of the beam report for one of both of DL reception and UL transmission. For example, the state can indicate a pair (x,y), where x is an ID as described in example B, and y takes a value from S, as described in example D. In example G, the state is regarding a combination of the MPE event, the panel entity, and the recommended usage of the beam report for one of both of DL reception and UL transmission. For example, the state can indicate a triple (x,y,z), where x is either (i) or (ii) in example A, y is an ID as described in example B, and z takes a value from S, as described in example D.

(167) Where 'none' implies no reporting of the recommended/preferred usage from the UE. Or 'none' implies that additional information about the state is not reported in the beam report.

(168) Regarding the beam metric for different values of the 'state', at least one of the following examples is used/configured. In one example, the beam metric can remain the same regardless of the reported value of the 'state'. In one example, the beam metric can be modified depending on the reported value of the 'state'. For example, the modification corresponds to a scaling of the beam metric L1-RSRP or L1-SINR. In one example, the beam metric can be m1 for when the reported value of the 'state' belongs to a set T1, and the beam metric can be m2 for when the reported value of the 'state' belongs to a set T2, where m1 and m2 are two different beam metrics, and T1 and T2 are two sets of possible values of the 'state'. The sets T1 and T2 can be disjoint (i.e., no common element). The (m1,m2) is according to at least one of the following examples: (m1,m2)=(L1-RSRP, scaled L1-RSRP) (m1,m2)=(L1-SINR, scaled SINR) (m1,m2)=(L1-RSRP, L1-SINR) (m1,m2)=(L1-RSRP, UL RSRP) (m1,m2)=(L1-RSRP, virtual PHR) (m1,m2)=(L1-SINR, UL RSRP) (m1,m2)=(L1-SINR, virtual PHR) In one example, the beam metric can be m1 for when the reported value of the 'state' belongs to a set T1, the beam metric can be m2 for when the reported value of the 'state' belongs to a set T2, and the beam metric can be m3 for when the reported value of the 'state' belongs to a set T3, where m1, m2, and m3 are three different beam metrics, and T1, T2, and T3 are three sets of possible values of the 'state'. The sets T1, T2, and T3 can be disjoint (i.e., no common element). The (m1,m2,m3) is according to at least one of the following examples: (m1,m2,m3)=(L1-RSRP, scaled L1-RSRP, UL RSRP) (m1,m2,m3)=(L1-RSRP, scaled L1-RSRP, virtual PHR) (m1,m2,m3)=(L1-SINR, scaled L1-SINR, UL RSRP) (m1,m2,m3)=(L1-SINR, scaled L1-SINR, virtual PHR) (m1,m2,m3)=(L1-RSRP, UL RSRP, virtual PHR) (m1,m2,m3)=(L1-SINR, UL RSRP, virtual PHR) (m1,m2,m3)=(L1-RSRP, L1-SINR, virtual PHR) (m1,m2,m3)=(L1-RSRP, L1-SINR, UL RSRP) In one example, the beam report can include a second metric (in addition to the beam metric) for some reported value(s) of the 'state'. That is, when the 'state' belongs to a set T1, the beam metric is reported (and the second metric is not reported), and when the 'state' belongs to a set T2, both the beam metric and the second metric are reported. Whether the second metric can be reported is determined based on at least one of the following examples. In one example, it is based on a configuration (e.g., via RRC and/or MAC CE and/or DCI). In one example, it is determined and reported by the UE. In this case, a two-part UCI (similar to Rel. 15 NR specification) can be used, wherein the UCI part 1 has an information about whether and/or the number of the second metric being reported in UCI part 2. In one example, it is subject to a UE capability reporting.

(169) In one example, Q.sub.1=1, indicating only one beam (resource indicator and metrics) is reported in the beam report. In this case, the 'state' value can be fixed, or reported by the UE, the details as explained above.

(170) In one example, when Q.sub.1>1, there are multiple beams (resource indicators and metrics) reported in the beam report. At least one of the following examples is used/configured regarding the 'state'. In one example, all Q.sub.1 beams correspond to the same value of the 'state'. For example, when the 'state' is related to the MPE, all Q.sub.1 beams are for (i) for MPE mitigation (taking MPE into account) or (ii) without MPE, i.e., are normal beams without taking MPE into account. A common bit field, IE, or parameter (e.g., 1-bit field or a parameter taking two values) is used for Q.sub.1 beams. In one example, N1 out of Q.sub.1 beams can correspond to one value of the 'state' (e.g., taking MPE into account) and remaining N2 beams can correspond to another value of the 'state' (e.g., normal beams without taking MPE into account). In one example, both N1 and N2 are fixed. In one example, N1 is fixed and N2 is configured (e.g., RRC and/or MAC CE and/or DCI). In one example, N1 is fixed or configured (e.g., RRC and/or MAC CE and/or DCI), and N2 is reported by UE. A 2-part UCI can be used. For example, the value N2 can be from 0 to Q.sub.1-N1. When N2=0, the UCI part 2 is absent not reported. When N2>1, one beam is reported via UCI part 2. Or UCI part 1 includes report for N1 beams, and a portion of report for N2 beams, and UCI part 2 includes remaining portion of report for N2 beams. In one example, both N1 and N2 reported by UE. A 2-part UCI can be used. For example, reported for a fixed number of beams is reported via UCI part 1, and the remaining via UCI part 2. Or UCI part 1 includes report for a portion of report Q.sub.1 beams, and UCI part 2 includes remaining portion of report for Q.sub.1 beams.

(171) In the rest of disclosure, the 'state' is assumed to indicate one of two values, whether MPE issue is detected by the UE or not. The embodiments and examples however are applicable other examples of the 'state' described above.

(172) In one sub-embodiment (I.2.1), the content or information conveyed by the 'state' is determined (or configured) according to at least

one of the following examples.

(173) In one example (I.2.1.1), the ‘state’ indicates one of two values, e.g., whether MPE issue is detected by the UE or not. For example, the information can be conveyed via a 1-bit field, where the field value=0 (or 1) indicates that the MPE issue is not detected, and the field value=1 (or 0) indicates that the MPE issue is detected. When the field value=0 (or 1), e.g., indicating that the MPE issue is not detected, the beam report includes beam indicator(s) (e.g., CRI(s)/SSBRI(s) and beam metric(s) (e.g., L1-RSRP, L1-SINR etc.) When the field value=1 (or 0), e.g., indicating that the MPE issue is detected, the beam report includes beam indicator(s) (e.g., CRI(s)/SSBRI(s) and beam metric(s) taking the event indicated by the field value into account, where a beam metric taking the event indicated by the field value can be at least one of the following: Example (I.2.1.1.1): the same as the beam metric when the field value=0, e.g., L1-RSRP, L1-SINR etc. Example (I.2.1.1.2): a modified beam metric, e.g., the modification corresponds to a scaling of the beam metric L1-RSRP or L1-SINR. Example (I.2.1.1.3): a new beam metric, e.g., UL RSRP, or virtual PHR. In one example, the UL RSRP is defined as $UL\ RSRP = L1-RSRP - P_{sub,DL} + P_{sub,UL}$, where L1-RSRP is the measured DL RSRP, $P_{sub,DL}$ is the DL transmit power, and $P_{sub,UL}$ is the available UL transmit power (or max UL transmit power). In one example, the virtual PHR is defined as a difference between the maximum available transmit power (e.g., $P_{sub,c,max}$) and the required transmit power.

(174) In one example (I.2.1.2), the information indicates whether MPE issue is detected by the UE or not. For example, the information can be conveyed via a parameter, where the parameter value=v0 indicates that the MPE issue is not detected, and the parameter value=v1 indicates that the MPE issue is detected.

(175) In one example (I.2.1.3), the information includes a value of the MPE. For example, the information can be conveyed via a M-bit field. When M=1, the 1-bit field indicates one of the two values {m.sub.0, m.sub.1}, where either both values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the other maps to the state “MPE detected and a corresponding (maximum or minimum) MPE value”. When M=2, the 2-bit field indicates one of the four values {m.sub.0, m.sub.1, m.sub.2, m.sub.3}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” In general, the M-bit field indicates one of the $2^{sup.M}$ values {m.sub.0, m.sub.1, . . . , m.sub.M-1}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” The set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1} can be fixed, or configured, e.g., via RRC and/or MAC CE and/or DCI. Or a maximum value for the MPE is configured, e.g., via RRC and/or MAC CE and/or DCI, and the set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1} is determined based on the configured maximum value. In one example, the set of MP values corresponds to the (maximum or minimum) power headroom (PHR) or Power Management Maximum Power Reduction (P-MPR) or UL duty-cycle values.

(176) In one example (I.2.1.4), the information includes a value of the MPE. For example, the information can be conveyed via a parameter taking a value from a set comprising M values. When M=2, the parameter indicates one of the two values {m.sub.0, m.sub.1}, where either both values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the other maps to the state “MPE detected and a corresponding (maximum or minimum) MPE value”. When M=4, the parameter indicates one of the four values {m.sub.0, m.sub.1, m.sub.2, m.sub.3}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” In general, the parameter indicates one of the M values {m.sub.0, m.sub.1, . . . , m.sub.M-1}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” The set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1} can be fixed, or configured, e.g., via RRC and/or MAC CE and/or DCI. Or a maximum value for the MPE is configured, e.g., via RRC and/or MAC CE and/or DCI, and the set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1} is determined based on the configured maximum value. In one example, the set of MP values corresponds to the (maximum or minimum) power headroom (PHR) or Power Management Maximum Power Reduction (P-MPR) or UL duty-cycle values.

(177) In one sub-embodiment (I.2.2), the reporting of the ‘state’ is determined (or configured) according to at least one of the following examples.

(178) In one example (I.2.2.1), the ‘state’ can always be included the beam report (i.e., reported by the UE).

(179) In one example (I.2.2.2), the ‘state’ can always be included the beam report (i.e., reported by the UE) when a condition is met, wherein, for example, the condition being that the UE is equipped with multiple antenna panels, or that the UE reports this (the UE-recommendation) as part of its UE capability, or that an event of interest can be detected by the UE.

(180) In one example (I.2.2.3), the reporting of the ‘state’ is configured via RRC and/or MAC CE (and/or triggered via DCI). When configured (or triggered), the UE always reports the ‘state’, otherwise, the UE does not report it. The UE can also be configured with a maximum value for the MPE. In one example, the configuration and/or triggering is UE-specific, in another example, the configuration and/or triggering is UE-group specific, in another example, the configuration and/or triggering is cell-specific.

(181) In one example (I.2.2.4), the reporting of the ‘state’ is reported by the UE without any configuration/triggering from the NW (i.e., the UE can initiate such reporting).

(182) In one example (I.2.2.5), the capability of reporting of the ‘state’ is reported by the UE (e.g., via UE capability reporting or as part of an UL transmission). Depending on the reported UE capability, the NW/gNB can configure (or trigger) the reporting of the ‘state’ via RRC and/or MAC CE (and/or DCI). The UE reports the ‘state’ according to the configuration/triggering from the NW/gNB.

(183) In one example (I.2.2.6), the UE decides (has the freedom to decide) whether the ‘state’ is included in the beam report, i.e., the UE may or may not include the ‘state’ in the beam report. For instance, such a decision can be made based on the event of interest. In one example, the beam report (with or without the ‘state’) is reported via a one-part UCI (similar to WB CSI reporting in Rel. 15 NR, the UE can append a fixed number of zeros in order to ensure that the payload remains the same regardless of whether the UE reports the ‘state’ or not). In one example, the beam report (with or without the ‘state’) is reported via a two-part UCI, wherein part 1 UCI includes an information whether the UE reports the ‘state’ or not. If the information indicates the ‘state’ being reported, it is reported via part 2 UCI.

(184) In one sub-embodiment (I.2.3), in addition to Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric) and ‘state’, the beam report can also include a UE-recommendation for UL TX beam selection/indication (by the NW/gNB), where the details about the UE-recommendation are as described in impediment I.1. Also, the ‘state’ and the UE-recommendation can either be reported jointly or separately.

(185) Upon receiving the beam report, the NW/gNB configures/indicates the UL TX beam indication (which can be the same as DL TX beam indication since beam correspondence holds) to the UE for UL transmission, where the beam indication indicates $N \geq 1$ UL TX beam(s). The beam indication can be via DL-TCI or UL-TCI or J-TCI (joint TCI) or other functionally equivalent entity such as

SpatialInfoInfo or SRI that is indicated via DCI and/or MAC CE and/or RRC. In one example $N=1$. In one example, $N=2$. Also, the beam indication can include a message (such as ACK or NACK/NULL), where the details about the message are as described in embodiment (I.1) and/or (I.4).

(186) In one sub-embodiment (I.2.4), a UE provides two sets of reports, either with a same report or separately in two different reports, wherein; A first report (sub-report) is a beam report including resource indicator(s) and possibly beam metrics, without reflecting the 'state'. A second report (sub-report) is a beam report including resource indicator(s) and possibly beam metrics, reflecting the 'state'. For example, the reported resource indicator(s) take into account the MPE effect. The beam metrics include the MPE effect.

(187) In one example, this behavior is configurable/triggered to the UE via RRC and/or MAC CE and/or DCI. When configured, the UE provides two reports (sub-reports), else it provides one report (the first report). In one example, the configuration and/or triggering is UE-specific, in another example, the configuration and/or triggering is UE-group specific, in another example, the configuration and/or triggering is cell-specific.

(188) In one embodiment (I.3), which is a variation of embodiment (I.2), except that the 'state' or indicator (e.g., indicating whether the beam report is for MPE or not) is not included in the beam report, it or beam reporting configuration is rather conveyed (provided) to the UE. In one example, this is provided via an RRC parameter. In one example, this RRC parameter is mpe-Reporting-r17 that allows the UE P-MPR-based MPE reporting (similar to Rel. 16 MPE reporting) included in the PHR report when a threshold is reached and is reported via (or triggered by) MAC-CE (entity), similar to the description in section 5.4.6 of TS 38/321. In one example, this RRC parameter is via a CSI request field (similar to the beam/CSI reporting configuration in Rel. 15/16). In one example, this RRC parameter is via a CSI request field (similar to the beam/CSI reporting configuration in Rel. 15/16), but its triggering is subject to (conditioned on) the MPE reporting included on the PHR report via MAC CE (similar to Rel. 16). That is, the beam reporting is triggered only when the UE reports MPE issue via the PHR report on MAC CE.

(189) In response to the configuration, the UE determines and reports a beam report based on the value of the 'state' or indicator it receives. The UL channel(s) or resource(s) configured for the beam report is according to at least one of the following: The beam report can be reported on PUCCH. If the PUCCH transmission that includes the beam report overlaps a PUSCH transmission, the uplink control information (UCI) with the beam report is multiplexed in the PUSCH. The beam report can be included in UCI transmitted on PUSCH, wherein, the PUSCH transmission can be one of: a PUSCH transmission scheduled by an UL grant, or a configured grant PUSCH transmission of Type 1 or of Type 2, or a Msg3 PUSCH transmission for random access procedure Type 1, or a MsgA PUSCH transmission for random access procedure Type 2. The beam report can be included in the PHR report and, is reported via MAC-CE or triggered via a MAC entity (scheduled to be reported via PUSCH carrying UL MAC). The beam report can be included in a combination of the PHR report and the UCI. That is a portion of the beam report is reported via MAC-CE (scheduled to be reported via PUSCH carrying UL MAC), and the remaining portion of the beam report is reported via the UCI (e.g., PUCCH, or PUSCH scheduled to carry UCI). In one example, P.sub.1 (P-MPR, SSBRI/CRI or UL TCI) pairs are reported via the PHR report, and the remaining Q.sub.1-P.sub.1 (P-MPR, SSBRI/CRI or UL TCI or an ID associated with antenna panel, e.g., CSI-RS/SRS resource set ID) pairs are reported via the UCI, when $Q_{sub.1} > P_{sub.1}$. In one example, $P_{sub.1}=1$. The maximum value of $Q_{sub.1}$ can be fixed to 4. The maximum value of $Q_{sub.1}$ can be reported by the UE via UE capability reporting. Alternatively, whether the UE supports the MPE reporting via the UCI is subject to (conditioned on) the UE capability. That is, only when the UE supports reporting on UCI, the UE can be configured with such reporting; otherwise (when the UE doesn't support), the UE can't be configured with MPE reporting on UCI, hence the MPE reporting can only be via MAC CE. In one example, a two-part PHR reporting via MAC CE is used (similar to two-part UCI reporting). A first part of the PHR reporting includes a fixed P.sub.1 (e.g., $P_{sub.1}=1$) number of beam report, and the second part of the PHR reporting includes the remaining $Q_{sub.1}-P_{sub.1}$ number of beam report. When the value $Q_{sub.1}$ is determined by the UE, then an information about the value $Q_{sub.1}-P_{sub.1}$ is included in the first part. Alternatively, whether the UE supports the two-part PHR reporting via the MAC CE is subject to (conditioned on) the UE capability. That is, only when the UE supports two-part PHR reporting, the UE can be configured with such reporting; otherwise (when the UE doesn't support), the UE can't be configured with two-part PHR reporting, hence the MPE reporting can only be via (one part) PHR reporting and/or UCI.

(190) In a variation, to facilitate MPE mitigation, the Rel-16 event-triggered P-MPR-based reporting (included in the PHR report when a threshold is reached, reported via MAC-CE) is extended as follows. In one example, $Q_{sub.1} \geq 1$ P-MPR values can be reported together with $M \geq 1$ SSBRI(s)/CRI(s), where the M SSBRI(s)/CRI(s) is selected by the UE from a candidate SSB/CSI-RS resource pool and/or from the pool of configured or activated TCI states (UL TCI or joint TCI). In one example, $Q_{sub.1} \geq 1$ P-MPR values can be reported together with $M \geq 1$ panel-associated indicator, e.g., an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID). In one example, $Q_{sub.1} \geq 1$ P-MPR values can be reported together with $M \geq 1$ SSBRI(s)/CRI(s) and $M \geq 1$ panel-associated indicator, e.g., an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID).

(191) In one example, $Q_{sub.1}=M$. In one example, $M > Q_{sub.1}$. In one example, $Q_{sub.1}=1$ and $M \geq 1$.

(192) In one example, $Q_{sub.1}$ represents the number of selected beams (via CRIs/SSBRIs) and/or the number of panels (via IDs associated with antenna panels).

(193) The supported values of $Q_{sub.1}$ can be fixed (e.g., 4). Or it can be configured from {1,2,3,4}. Or the max value of $Q_{sub.1}$ is reported by the UE (e.g., via UE capability reporting), and the configured value is less or equal to the reported value.

(194) As explained above, this reporting can also be via a combination of PHR reporting and the UCI reporting, or via a two-part PHR reporting. The additional reporting quantities can also be included, which can comprise UL RSRP, virtual PHR, DL RSRP etc., details of which are according to at least one example in this disclosure.

(195) FIG. 15 illustrates an example of single-entry MAC CE for multiple MPE reports **1500** according to embodiments of the present disclosure. The embodiment of the single-entry MAC CE for multiple MPE reports **1500** illustrated in FIG. 15 is for illustration only. FIG. 15 does not limit the scope of this disclosure to any particular implementation of the single-entry MAC CE for multiple MPE reports **1500**.

(196) FIG. 16 illustrates an example of single-entry MAC CE for multiple MPE reports **1600** according to embodiments of the present disclosure. The embodiment of the single-entry MAC CE for multiple MPE reports **1600** illustrated in FIG. 16 is for illustration only. FIG. 16 does not limit the scope of this disclosure to any particular implementation of the single-entry MAC CE for multiple MPE reports **1600**.

(197) For beam reporting included in the PHR reporting via MAC CE, two examples of the single-entry MAC CE for multiple MPE reports are shown in FIG. 15 and FIG. 16.

(198) In one example (I.3.1), the content of the beam report is according to one of the examples in this disclosure (cf. embodiment (I.1)/(I.2)). A few similar or other examples of the content of the beam report are as follows.

(199) In one example (I.3.1.0), the beam report includes $Q_{sub.1}$ pairs UP, where I=indicator=SSBRI or CRI or TCI (UL TCI or joint-TCI) or an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID), and J=metric=m=a MPE metric when there is an MPE issue

(MPE issue detected). In one example, the metric is P-MPR. In one example, the metric is UL RSRP. In one example, the metric is a combination of DL-RSRP and P-MPR. In one example, the metric is a combination of DL-RSRP and virtual PHR. The details of these metrics are according to an example in this disclosure (e.g., example (I.3.1.1)).

(200) In one example (I.3.1.1), the beam report includes Q.sub.1 pairs (I, J), where I=indicator=SSBRI or CRI or TCI (UL TCI or joint-TCI) or an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID), and J=metric=m is one of m1 and m2 m=m1, when 'state' or indicator=0 or v0, which for example indicates a normal metric when there is no MPE issue (MPE issue not detected), m=m2, when 'state' or indicator=1 or v1, which for example indicates a MPE metric when there is an MPE issue (MPE issue detected).

(201) At least one of the following examples is used/configured for (m1, m2). In one example, (m1, m2)=(DL RSRP, UL RSRP), where DL RSRP=L1-RSRP as in Rel.15 or a modified L1-RSRP, where the modification corresponds to one of the following. a scaling of the beam metric by ζ , i.e., DL RSRP $\times\zeta$; where ζ can depend on path loss subtracting (reducing) a factor γ , i.e., DL RSRP $-\gamma$; where γ can depend on path loss UL RSRP=DL RSRP-P-MPR or UL RSRP=L1-RSRP-P.sub.DL+P.sub.UL as defined in example (I.2.1.1.3). In one example, (m1, m2)=(DL SINR, UL RSRP), where DL RSRP=L1-SINR as in Rel.16 or a modified L1-SINR, where the modification corresponds to one of the following. a scaling of the beam metric by ζ , i.e., DL SINR $\times\zeta$; where ζ can depend on path loss subtracting (reducing) a factor γ , i.e., DL SINR $-\gamma$; where γ can depend on path loss UL RSRP=DL RSRP-P-MPR or UL RSRP=L1-RSRP-P.sub.DL+P.sub.UL as defined in example (I.2.1.1.3). In one example, (m1, m2)=(DL RSRP, vPHR), where DL RSRP is as explained in above examples, and vPHR is the virtual PHR as defined in this disclosure [REF8].

(202)

$$vPHR_{b,f,c}(i,j,q_d,l) = P_{CMAX,f,c}(i,q_d) - \{P_{OPUSCH,b,f,c}(i) + 10\log_{10}(2^{ref} M_{RB,b,f,c}^{ref,PUSCH}(i) + b_{f,c}(j)PL_{b,f,c}(q_d) + TF_{ref,b,f,c} + f_{b,f,c}(i,l)\} [dB]$$

In one example, (m1, m2)=(DL SINR, vPHR), where DL SINR and vPHR are as explained in above examples. In one example, (m1, m2)=(DL RSRP, UL RSRP or vPHR), where m2=UL RSRP or vPHR based on configuration (e.g., RRC via CSI-AperiodicTriggerState or new parameter), or codepoint (CSI request field in DCI or a separate field in DCI) or based on UE capability. The UE may support only one of the two metrics, or UE may support both, and the NW will configure/trigger one of the two metrics based on the UE capability reporting. The details of the three metrics are as explained in above examples. In one example, (m1, m2)=(DL SINR, UL RSRP or vPHR), where m2=UL RSRP or vPHR based on configuration (e.g., RRC via CSI-AperiodicTriggerState or new parameter), or codepoint (CSI request field in DCI or a separate field in DCI) or based on UE capability. The UE may support only one of the two metrics, or UE may support both, and the NW will configure/trigger one of the two metrics based on the UE capability reporting. The details of the three metrics are as explained in above examples. In one example, (m1, m2)=one of the examples in embodiment (I.2).

(203) In one example (I.3.1.1A), the beam report includes Q.sub.1 pairs (I,J), where I=indicator=SSBRI or CRI or TCI (UL TCI or joint-TCI) or an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID), and J=metric=m is one of m1 and a pair (m2, m3) m=m1, when 'state' or indicator=0 or v0, which for example indicates a normal metric when there is no MPE issue (MPE issue not detected), m=a pair (m2, m3), when 'state' or indicator=1 or v1, which for example indicates a MPE metric when there is an MPE issue (MPE issue detected).

(204) At least one of the following examples is used/configured for (m1, m2, m3). In one example, m1=DL RSRP, and (m2, m3)=(DL RSRP, P-MPR), where DL RSRP is as explained in above examples, P-MPR (Power Management-Maximum Power Reduction) is the reduction in maximum UL power due to the maximum permissible exposure (MPE) regulations. In one example, m1=DL RSRP, and (m2, m3)=(UL RSRP, P-MPR), where DL RSRP, UL RSRP, and P-MPR are as explained in above examples. In one example, m1=DL RSRP, and (m2, m3)=(DL RSRP, vPHR), where DL RSRP and vPHR are as explained in above examples. In one example, m1=DL RSRP, and (m2, m3)=(UL RSRP, vPHR), where DL RSRP, UL RSRP, and vPHR are as explained in above examples. In one example, m1=DL RSRP, and (m2, m3)=(DL RSRP, UL RSRP), where DL RSRP and UL RSRP are as explained in above examples. In one example, m1=DL RSRP, and (m2, m3)=(P-MPR, vPHR), where P-MPR and vPHR are as explained in above examples. In one example, m1=DL SINR, and (m2, m3) is according to one of the above examples.

(205) Table 1, Table 2, and Table 3 define three examples of the P-MPR report mapping.

(206) TABLE-US-00002 TABLE 1 Mapping of P-MPR Reported value Measured quantity value Unit P-MPR_00 3 □ PMP-R < 6 dB P-MPR_01 6 □ PMP-R < 9 dB P-MPR_02 9 □ PMP-R < 12 dB P-MPR_03 PMP-R ≥ 12 dB

(207) TABLE-US-00003 TABLE 2 Mapping of P-MPR Reported value Measured quantity value Unit P-MPR_00 1.5 □ PMP-R < 3 dB P-MPR_01 3 □ PMP-R < 4.5 dB P-MPR_02 4.5 □ PMP-R < 6 dB P-MPR_03 6 □ PMP-R < 7.5 dB P-MPR_04 7.5 □ PMP-R < 9 dB P-MPR_05 9 □ PMP-R < 10.5 dB P-MPR_06 10.5 □ PMP-R < 12 dB P-MPR_07 PMP-R ≥ 12 dB

(208) TABLE-US-00004 TABLE 3 Mapping of P-MPR Reported value Measured quantity value Unit P-MPR_00 2 □ PMP-R < 4 dB P-MPR_01 4 □ PMP-R < 6 dB P-MPR_02 6 □ PMP-R < 8 dB P-MPR_03 8 □ PMP-R < 10 dB P-MPR_04 10 □ PMP-R < 12 dB P-MPR_05 12 □ PMP-R < 14 dB P-MPR_06 14 □ PMP-R < 16 dB P-MPR_07 PMP-R ≥ 16 dB

(209) When Q.sub.1>1, the first P-MPR can be reported according to one of Table 1 through Table 3, and the remaining Q.sub.1-1 P-MPR can be reported according to at least one of the following. In one example, the P-MPR values are reported without any ordering. Hence, the same payload (number of bits) is used for reporting each P-MPR value. In one example, the P-MPR values are ordered (sorted) in decreasing order and reported in a differential manner. That is, the first P-MPR (v.sub.0) is reported using x.sub.0 bits (e.g., x.sub.0=2 or 3) and acts a reference for the second P-MPR. The differential second P-MPR (d.sub.1) is reported using x.sub.1 bits (e.g., x.sub.1=1 or 2 bits), the actual second P-MPR is given by v.sub.1=d.sub.1+v.sub.0, and the second P-MPR acts a reference for the third P-MPR, and so on. Alternatively, the first P-MPR (v.sub.0) acts a reference for all of the remaining P-MPR values (second, third, and so on). In one example, the P-MPR values are reported in a differential manner. That is, the first P-MPR (v.sub.0) is reported using x.sub.0 bits (e.g., x.sub.0=2 or 3) and acts a reference for the second P-MPR. The differential second P-MPR (d.sub.1) is reported using x.sub.1 bits (e.g., x.sub.1=1 or 2 bits), the actual second P-MPR is given by v.sub.1=d.sub.1+v.sub.0, and the second P-MPR acts a reference for the third P-MPR, and so on. Alternatively, the first P-MPR (v.sub.0) acts a reference for all of the remaining P-MPR values (second, third, and so on).

(210) In one example (I.3.1.2), the beam report includes Q.sub.1 pairs UP, where I=indicator=SSBRI or CRI or TCI (UL TCI or joint-TCI) or an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID), and J=metric=m is one of m1 and m2 m=m1, when 'state' or indicator=0 or v0, which for example indicates a first MPE metric when there is MPE issue (MPE issue detected), m=m2, when 'state' or indicator=1 or v1, which for example indicates a second MPE metric when there is MPE issue (MPE issue detected).

(211) At least one of the following examples is used/configured for (m1, m2). In one example, (m1, m2)=(vPHR, UL RSRP), where UL RSRP and vPHR are as described above. In one example, (m1, m2)=one of the examples in embodiment (I.2).

(212) In one example (I.3.1.3), the beam report includes Q.sub.1 pairs UP, where I=indicator=SSBRI or CRI or TCI (UL TCI or joint-TCI) or an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID), and J=metric=m is one of m1, m2, and m3 m=m1, when 'state' or indicator=0 or v0, which for example indicates a normal metric when there is no MPE issue (MPE issue not detected), m=m2, when

'state' or indicator=1 or v1, which for example indicates a first MPE metric when there is MPE issue (MPE issue detected), m=m3, when 'state' or indicator=2 or v2, which for example indicates a second MPE metric when there is MPE issue (MPE issue detected).

(213) At least one of the following examples is used/configured for (m1, m2, m3). In one example, (m1, m2, m3)=(DL RSRP, UL RSRP, vPHR), where DL RSRP, UL RSRP, and vPHR are as described above. In one example, (m1, m2, m3)=(DL SINR, UL RSRP, vPHR), where DL RSRP, UL RSRP, and vPHR are as described above. In one example, (m1, m2)=one of the examples in embodiment (I.2).

(214) In one example (I.3.1.3A), the beam report includes Q.sub.1 pairs (I,J), where I=indicator=SSBRI or CRI or TCI (UL TCI or joint-TCI) or an ID associated with antenna panel (e.g., SRS/CSI-RS resource set ID), and J=metric=m is one of m1, m2, and a pair (m3, m4) m=m1, when 'state' or indicator=0 or v0, which for example indicates a normal metric when there is no MPE issue (MPE issue not detected), m=m2, when 'state' or indicator=1 or v1, which for example indicates a MPE metric when there is MPE issue (MPE issue detected), m=a pair (m3, m4), when 'state' or indicator=2 or v2, which for example indicates a MPE metric pair when there is MPE issue (MPE issue detected).

(215) At least one of the following examples is used/configured for (m1, m2, m3, m4). In one example, m1=DL RSRP, m2=UL RSRP, and (m3, m4) is one of the examples for (m2, m3) in example (I.3.1.1.A). In one example, m1=DL RSRP, m2=vPHR, and (m3, m4) is one of the examples for (m2, m3) in example (I.3.1.1.A). In one example, m1=DL RSRP, m2=P-MPR, and (m3, m4) is one of the examples for (m2, m3) in example (I.3.1.1.A). In one example, m1=DL SINR, m2=UL RSRP, and (m3, m4) is one of the examples for (m2, m3) in example (I.3.1.1.A). In one example, m1=DL SINR, m2=vPHR, and (m3, m4) is one of the examples for (m2, m3) in example (I.3.1.1.A). In one example, m1=DL SINR, m2=P-MPR, and (m3, m4) is one of the examples for (m2, m3) in example (I.3.1.1.A).

(216) In one example, the UE reports via its capability reporting whether it supports one of or both of the MPE metrics, m=m2 and m=(m3, m4).

(217) In one example (I.3.2), the state or indicator (e.g., 1-bit or 2-bit indicator) is conveyed to the UE explicitly, and the UE determines/reports the beam report according to the received state or indicator.

(218) In one example (I.3.2.1), the state or indicator is conveyed via a DCI. For example, a CSI request field or a separate field in the DCI can be used for this purpose.

(219) In one example (I.3.2.2), the state or indicator is conveyed via MAC CE. For example, an activation command that activates CSI-RS subset selection or a separate activation command can be used for this purpose.

(220) In one example (I.3.2.3), the state or indicator is conveyed via higher layer (RRC). For example, a trigger state definition (CSI-AperiodicTriggerState) for CSI reporting or a separate RRC parameter can be used for this purpose.

(221) FIG. 17 illustrates an example of single-entry MAC CE for multiple MPE reports **1700** according to embodiments of the present disclosure. The embodiment of the single-entry MAC CE for multiple MPE reports **1700** illustrated in FIG. 17 is for illustration only. FIG. 17 does not limit the scope of this disclosure to any particular implementation of the single-entry MAC CE for multiple MPE reports **1700**.

(222) In one example (I.3.3), the beam reporting is "triggering" implicitly (without CSI configuration/triggering). For example, a UE-initiated or event-triggered mechanism can be used as the trigger. For instance, the implicit trigger is based on the MPE reporting via single or multiple entry PHR reporting on a MAC entity (cf. Section 5.4.6, 6.1.3.8/9, TS 38.321) when the MPE reporting includes P=1 and the corresponding MPE (P-PMR) value (FIG. 13 for single entry PHR MAC CE). That is, when a UE reports the MPE value and P=1, the UE will report the beam report, either in the same time slot (as for the PHR reporting, jointly or separately from the PHR reporting) or possibly after at least a time delay of N time slots (OFDM symbols) from the time slot (or OFDM symbols) which contains the MPE reporting, where N can be determined according to at least one of the following examples. In one example (I.3.3.1), N is fixed (pre-determined), and not configured, e.g., N=0 or 4. In one example (I.3.3.2), N is the earliest time slot scheduled for UL transmission (via configured grant type 1 or 2, or via DCI). In one example (I.3.3.3), N is configured via higher layer (RRC, TriggerState or separate parameter), or MAC CE (MAC CE that performs subset selection of aperiodic CSI-RS resources or a separate MAC CE), or DCI (CSI request field or a separate field). In one example (I.3.3.4), N or the minimum value of N or the maximum value of N is reported by the UE as part of the UE capability. In one example (I.3.3.5), when the subcarrier spacings (SCSs) of the DL and UL channels and/or BWPs are different, the value of N can depend on the SCS of the active UL BWP and/or UL channel used for beam reporting (e.g., PUSCH/PUCCH/PRACH) and/or active DL BWP. For example, the time slot (or OFDM symbols) for UL transmission can be determined based on a scaling factor x or $\lfloor x \rfloor$ or $\lceil x \rceil$, where $x = 2 \cdot \sup. \mu \cdot \text{sub}.1 / 2 \cdot \sup. \mu \cdot \text{sub}.2 = 2 \cdot \sup. \mu \cdot \text{sub}.1 \cdot \sup. - \mu \cdot \text{sub}.2$, and $\mu \cdot \text{sub}.1$ and $\mu \cdot \text{sub}.2$ respectively indicate SCSs of DL and UL channels and/or BWPs taking a value from {0, 1, 2, 3, 4} indicating {15, 30, 60, 120, 240} kHz SCSs, respectively.

(223) The UL transmission including the beam report is via a configured grant PUSCH resource and starts at the earliest configured grant PUSCH opportunity after N time slots from the time slot (or OFDM symbols) which contains the MPE reporting. Or the UL transmission including the beam report is via a configured PUCCH resource (configured grant Type 1 or Type 2) and starts at the earliest configured PUCCH opportunity after N time slots from the time slot (or OFDM symbols) which contains the MPE reporting. Or the UL transmission including the beam report is via a PRACH (type 1 or type), wherein the preamble starts at the earliest PRACH opportunity after N time slots from the time slot (or OFDM symbols) which contains the MPE reporting.

(224) In one example (I.3.4), the beam reporting can be "triggered" (cf. example (I.3.2)) explicitly only when (after) the UE reports the MPE reporting (cf. example (I.3.3)), for example, when the UE reports the MPE reporting via the PHR reporting on a MAC entity. That is, the CSI request field in DCI can trigger the beam reporting only when the NW receives the MPE reporting via the PHR reporting on a MAC entity from the UE.

(225) In one embodiment (I.4), the RS(s) for SSBRI/CRI reporting is (are) according to at least one of the following examples: In one example (I.4.1), the RS(s) for SSBRI/CRI reporting is (are) higher layer configured measurement RS resources as in Rel.15/16 beam measurement/reporting. For aperiodic CSI-RS(s), a MAC CE based subset selection can be used when the number of higher layer configured aperiodic CSI-RS resources are large than a threshold (determined based on the bit width of the DCI codepoint/field, CSI request, that triggers aperiodic CSI-RS measurement and reporting. In one example (I.4.2), the RS(s) for SSBRI/CRI reporting is (are) the QCL source RS(s) included in the active TCI state(s) that are activated via MAC CE. In one example (I.4.3), the RS(s) for SSBRI/CRI reporting is (are) the QCL source RS(s) for the higher layer (RRC) configured TCI state(s). In one example (I.4.4), the RS(s) for SSBRI/CRI reporting is (are) the RS(s) configured as PL-RS(s). In one example (I.4.5), the RS(s) for SSBRI/CRI reporting is (are) a combination of two sets of RSs according to example (I.4.x) and example (I.4.y), where $x \neq y$ and x and y belong to {1,2,3,4}.

(226) In one embodiment (I.5), the UE is configured with a beam reporting that doesn't include any UE-recommendation, i.e., the beam report includes Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), where $Q \cdot \text{sub}.1 \leq P \cdot \text{sub}.1$ as described in embodiment (I.1). In one example, $Q \cdot \text{sub}.1 = 1$. In one example, $Q \cdot \text{sub}.1$ is configured via RRC and/or MAC CE. The resource indicator indicates a DL measurement RS resource index from the $P \cdot \text{sub}.1$ DL measurement RS resources. Examples of resource indicator include CRI (when DL measurement RS is CSI-RS) and SSB-RI (when DL measurement RS is SSB).

(227) In one example (I.5.1), the beam metric is represented by the channel (DL or UL channel since beam correspondence holds). At least one of the following examples is used for the beam metric. In example (I.5.1.1), the beam metric is L1-RSRP, which corresponds to DL RSRP with or without any modification; where the modification corresponds to one of the following. a scaling of the beam metric by ζ , i.e., DL RSRP $\times\zeta$; where ζ can depend on path loss subtracting (reducing) a factor γ , i.e., DL RSRP $-\gamma$; where γ can depend on path loss. In example (I.5.1.2), the beam metric is L1-SINR, which corresponds to DL SINR with or without any modification; where the modification corresponds to one of the following. a scaling of the beam metric by ζ , i.e., DL SINR $\times\zeta$; where ζ can depend on path loss subtracting (reducing) a factor γ , i.e., DL SINR $-\gamma$; where γ can depend on path loss. In example (I.5.1.3), the beam metric is CQI. In example (I.5.1.4), the beam metric is hypothetical BLER (block error rate). In example (I.5.1.5), the beam metric is virtual power head room (PH), which corresponds to the difference between the maximum available transmit power (e.g., P.sub.c,max) and the required transmit power, as defined in [REF8], copied below.

(228)

$$vPH_{b,f,c}(i,j,q_d,l) = P_{CMAX,f,c}(i,q_d) - \{P_{OPUSCH,b,f,c}(i) + 10\log_{10}(2^{-ref} M_{RB,b,f,c}^{ref,PUSCH}(i)) + b_{f,c}(j)PL_{b,f,c}(q_d) + TF_{ref,b,f,c} + f_{b,f,c}(i,l)\}[\text{dB}]$$

In example (I.5.1.6), the beam metric is P-MPR (Power Management–Maximum Power Reduction), which corresponds to the reduction in maximum UL power due to the maximum permissible exposure (MPE) regulations. In example (I.5.1.7), the beam metric is UL RSRP, which corresponds to L1-RSRP–P.sub.DL+P.sub.UL, where L1-RSRP is the measured DL RSRP, P.sub.DL is the DL transmit power, and P.sub.UL is the available UL transmit power (or max UL transmit power).

(229) The payload (number of bits) of the beam metric can be the same (e.g., 6 or 7 bits) regardless of the type of the beam metric. Or the payload can vary depending on the type of the beam metric.

(230) In example (I.5.2), when Q.sub.1 $\geq$ 1, the beam metrics can be reported according to at least one of the following. In example (I.5.2.1), the beam metrics are of the same type for all Q.sub.1 pairs of UM, where the type of the beam metric is either fixed (e.g., one of example (I.5.1.1) through (I.5.1.7)) or configured (e.g., one example (I.5.1.1) through (I.5.1.7) via RRC or MAC CE or DCI) or reported by the UE (e.g., via UE capability or together with the beam report or separate from the beam report). In example (I.5.2.2), the beam metrics can be one of the two types from example (I.5.1.a) and (I.5.1.b). In one example (a,b) is fixed, e.g., (a,b)=(1,5), or configured (via RRC or MAC CE or DCI), or reported by the UE (e.g., via UE capability or together with the beam report or separate from the beam report). Let N.sub.a and N.sub.b be number of beam reports of types a and b, respectively. Then, N.sub.a and N.sub.b can be fixed or configured (via RRC or MAC CE or DCI), or reported by the UE (e.g., via UE capability or together with the beam report or separate from the beam report). The location (indices) of these beam metrics can be fixed, e.g., first N.sub.a metrics are of type a, and the rest of type b. Or the location (indices) of these beam metrics can be configured (via RRC or MAC CE or DCI), or reported by the UE (e.g., via UE capability or together with the beam report or separate from the beam report).

(231) In example (I.5.3), when Q.sub.1 $>$ 1, the beam metrics can be reported according to at least one of the following. In example (I.5.3.1), the beam metrics are reported without any ordering. Hence, the same payload (number of bits) is used for reporting each beam metric. In example (I.5.3.2), the beam metrics are ordered (sorted) in decreasing order and reported in a differential manner. That is, the first beam metric (v.sub.0) is reported using x.sub.0 bits (e.g., x.sub.0=6 or 7) and acts a reference for the second beam metric. The differential second beam metric (d.sub.1) is reported using x.sub.1 bits (e.g., x.sub.1=4 or 3 bits), the actual second beam metric is given by v.sub.1=d.sub.1+v.sub.0, and the second beam metric acts a reference for the third beam metric, and so on. Alternatively, the first beam metric (v.sub.0) acts a reference for all of the remaining beam metrics (second, third, and so on). In example (I.5.3.2A), the beam metrics are reported in a differential manner. That is, the first beam metric (v.sub.0) is reported using x.sub.0 bits (e.g., x.sub.0=6 or 7) and acts a reference for the second beam metric. The differential second beam metric (d.sub.1) is reported using x.sub.1 bits (e.g., x.sub.1=4 or 3 bits), the actual second beam metric is given by v.sub.1=d.sub.1+v.sub.0, and the second beam metric acts a reference for the third beam metric, and so on. Alternatively, the first beam metric (v.sub.0) acts a reference for all of the remaining beam metrics (second, third, and so on). In example (I.5.3.3), when the beam metrics can be one of the two types (cf. example (I.5.2.2)), then the beam metrics can be reported according to at least one of the following. In one example, both types of the beam metric are reported according to example (I.5.3.1). In one example, the first type (a) of the beam metric is reported according to example (I.5.3.1) and the second type (b) of the beam metric is reported according to example (I.5.3.2). In one example, the second type (a) of the beam metric is reported according to example (I.5.3.1) and the first type (b) of the beam metric is reported according to example (I.5.3.2). In one example, both types of the beam metric are reported according to example (I.5.3.2).

(232) When the beam metric is virtual power headroom (vPH), the bitwidth for CRI, SSBRI, vPH, and differential vPH are provided in Table 4.

(233) TABLE-US-00005 TABLE 4 CRI, SSBRI, and RSRP Field Bitwidth CRI [log.sub.2(K.sub.s.sup.CSI–RS)] SSBRI [log.sub.2(K.sub.s.sup.SSB)] vPH 6 or 7 Differential vPH 4

where K.sub.S.sup.CSI–RS is the number of CSI-RS resources in the corresponding resource set, and K.sub.S.sup.SSB is the configured number of SS/PBCH blocks in the corresponding resource set for reporting ‘ssb-Index-vPH’.

(234) When the bitwidth of vPH reporting is 6 bits, the vPH reporting range is from –32 . . . +38 dB. Table 5 defines the report mapping.

(235) TABLE-US-00006 TABLE 5 Power headroom report mapping Reported value Measured quantity value (dB) vPH_0 PH < –32 vPH_1 –32 $\leq$ PH < –31 vPH_2 –31 $\leq$ PH < –30 vPH_3 –30 $\leq$ PH < –29 vPH_53 20 $\leq$ PH < 21 vPH_54 21 $\leq$ PH < 22 vPH_55 22 $\leq$ PH < 24 vPH_56 24 $\leq$ PH < 26 vPH_57 26 $\leq$ PH < 28 vPH_58 28 $\leq$ PH < 30 vPH_59 30 $\leq$ PH < 32 vPH_60 32 $\leq$ PH < 34 vPH_61 34 $\leq$ PH < 36 vPH_62 36 $\leq$ PH < 38 vPH_63 PH $\geq$ 38

(236) When the bitwidth of vPH reporting is 7 bits, the vPH reporting range is from –32 . . . +38 dB. Table 6 defines the report mapping.

(237) TABLE-US-00007 TABLE 6 Power headroom report mapping Reported value Measured quantity value (dB) vPH_0 PH < –32 vPH_1 –32 $\leq$ PH < –31.5 vPH_2 –31.5 $\leq$ PH < –31 vPH_3 –31 $\leq$ PH < –30.5 vPH_4 –30.5 $\leq$ PH < –30 vPH_109 22 $\leq$ PH < 22.5 vPH_110 22.5 $\leq$ PH < 23 vPH_111 23 $\leq$ PH < 23.5 vPH_112 23.5 $\leq$ PH < 24 vPH_113 24 $\leq$ PH < 25 vPH_114 25 $\leq$ PH < 26 vPH_125 36 $\leq$ PH < 37 vPH_126 37 $\leq$ PH < 38 vPH_127 PH $\geq$ 38

(238) When the bitwidth of differential vPH reporting is 4 bits, the reporting range of differential SS-vPH (differential vPH associated with SSB block) and CSI-vPH (differential vPH associated with CSI-RS) for L1 reporting is defined from 0 dBm to –30 dB with x dB resolution. Examples of the mapping of measured quantity are defined in Table 4 and Table 7 for x=2 and in Table 5 and Table 6 for x=1. The range in the signaling may be larger than the guaranteed accuracy range.

(239) TABLE-US-00008 TABLE 7 Differential SS-vPH and CSI-vPH measurement (for L1 reporting) report mapping Measured quantity value (difference in measured vPH Reported value from the reference vPH) Unit DIFFvPH_0 0 $\geq$ Δ vPH > –2 dB DIFFvPH_1 –2 $\geq$ Δ vPH > –4 dB DIFFvPH_2 –4 $\geq$ Δ vPH > –6 dB DIFFvPH_3 –6 $\geq$ Δ vPH > –8 dB DIFFvPH_4 –8 $\geq$ Δ vPH > –10 dB DIFFvPH_5 –10 $\geq$ Δ vPH > –12 dB DIFFvPH_6 –12 $\geq$ Δ vPH > –14 dB DIFFvPH_7 –14 $\geq$ Δ vPH > –16 dB DIFFvPH_8 –16 $\geq$ Δ vPH > –18 dB DIFFvPH_9 –18 $\geq$

$\Delta vPH > -20$ dB DIFFvPH_10 $-20 \geq \Delta vPH > -22$ dB DIFFvPH_11 $-22 \geq \Delta vPH > -24$ dB DIFFvPH_12 $-24 \geq \Delta vPH > -26$ dB DIFFvPH_13 $-26 \geq \Delta vPH > -28$ dB DIFFvPH_14 $-28 \geq \Delta vPH > -30$ dB DIFFvPH_15 $-30 \geq \Delta vPH$ dB
(240) TABLE-US-00009 TABLE 8 Differential SS-vPH and CSI-vPH measurement (for L1 reporting) report mapping Measured quantity value (difference in measured vPH Reported value from the reference vPH) Unit DIFFvPH_0 $0 \geq \Delta vPH > -1$ dB DIFFvPH_1 $-1 \geq \Delta vPH > -2$ dB DIFFvPH_2 $-2 \geq \Delta vPH > -3$ dB DIFFvPH_3 $-3 \geq \Delta vPH > -4$ dB DIFFvPH_4 $-4 \geq \Delta vPH > -5$ dB DIFFvPH_5 $-5 \geq \Delta vPH > -6$ dB DIFFvPH_6 $-6 \geq \Delta vPH > -7$ dB DIFFvPH_7 $-7 \geq \Delta vPH > -8$ dB DIFFvPH_8 $-8 \geq \Delta vPH > -9$ dB DIFFvPH_9 $-9 \geq \Delta vPH > -10$ dB DIFFvPH_10 $-10 \geq \Delta vPH > -11$ dB DIFFvPH_11 $-11 \geq \Delta vPH > -12$ dB DIFFvPH_12 $-12 \geq \Delta vPH > -13$ dB DIFFvPH_13 $-13 \geq \Delta vPH > -14$ dB DIFFvPH_14 $-14 \geq \Delta vPH > -15$ dB DIFFvPH_15 $-15 \geq \Delta vPH$ dB
(241) TABLE-US-00010 TABLE 9 Differential SS-vPH and CSI-vPH measurement (for L1 reporting) report mapping Measured quantity value (difference in measured vPH Reported value from the reference vPH) (dB) DIFFvPH_0 $\Delta vPH < -7$ DIFFvPH_1 $-7 \leq \Delta vPH < -6$ DIFFvPH_2 $-6 \leq \Delta vPH < -5$ DIFFvPH_3 $-5 \leq \Delta vPH < -4$ DIFFvPH_7 $-1 \leq \Delta vPH < 0$ DIFFvPH_8 $0 \leq \Delta vPH < 1$ DIFFvPH_14 $6 \leq \Delta vPH < 7$ DIFFvPH_15 $7 \leq \Delta vPH$
(242) TABLE-US-00011 TABLE 10 Differential SS-vPH and CSI-vPH measurement (for L1 reporting) report mapping Measured quantity value (difference in measured vPH Reported value from the reference vPH) (dB) DIFFvPH_0 $\Delta vPH < -14$ DIFFvPH_1 $-14 \leq \Delta vPH < -12$ DIFFvPH_2 $-12 \leq \Delta vPH < -10$ DIFFvPH_3 $-10 \leq \Delta vPH < -8$ DIFFvPH_7 $-2 \leq \Delta vPH < 0$ DIFFvPH_8 $0 \leq \Delta vPH < 2$ DIFFvPH_14 $12 \leq \Delta vPH < 14$ DIFFvPH_15 $14 \leq \Delta vPH$
(243) In one example, the reference vPH corresponds to the first vPH (associated with the first SSBRI/CRI with index i=0). In one example, for i>1, the reference vPH for SSBRI/CRI index i is the (i-1)-th vPH (associated with the previous SSBRI/CRI with index i-1).
(244) In one example, vPH and differential vPH reporting is (using) based on Table 5 and 7, respectively.
(245) In one example, vPH and differential vPH reporting is (using) based on Table 6 and 7, respectively.
(246) In one example, vPH and differential vPH reporting is (using) based on Table 5 and 8, respectively.
(247) In one example, vPH and differential vPH reporting is (using) based on Table 6 and 8, respectively.
(248) In one example, vPH and differential vPH reporting is (using) based on Table 5 and 9, respectively.
(249) In one example, vPH and differential vPH reporting is (using) based on Table 6 and 9, respectively.
(250) In one example, vPH and differential vPH reporting is (using) based on Table 5 and 10, respectively.
(251) In one example, vPH and differential vPH reporting is (using) based on Table 6 and 10, respectively.
(252) The P.sub.CMAX,f,c(i, q.sub.d) to calculate vPH (and differential vPH) takes into account the P-MPR due to MPE regulations, as described above. The pathloss PL.sub.b,f,c(q.sub.d) to calculate vPH (and differential vPH) can be based on L1-RSRP measured from the corresponding SSB/CSI-RS. The beam reporting via the CSI reporting (higher layer CSI-ReportConfig) can be configured/initiated based on the UE triggered-event on MPE reporting via MAC CE entity (cf. Section 5.4.6, TS 38.321).
(253) In one example (I.5.4), the value of Q.sub.1 can be determined according to at least one of the following. In one example (I.5.4.1), the value of Q.sub.1 is fixed, e.g., Q.sub.1=1 or 2. In one example (I.5.4.2), the value of Q.sub.1 is configured (via RRC or MAC CE or DCI). In one example (I.5.4.3), the maximum value of Q.sub.1 or the set of (candidate) values of Q.sub.1 is reported by the UE as part of UE capability reporting. The configured value of Q.sub.1 is subject to the UE capability reporting. In one example (I.5.4.4), the value of Q.sub.1 is reported by the UE (as part of the beam reporting). In one example (I.5.4.5), the maximum value of Q.sub.1 can be fixed, e.g., 4. The value of Q.sub.1 can be configured (e.g., in higher layer CSI-reportConfig configuring beam reporting), e.g., from {1,2,3,4}. This configuration is subject to the UE capability reporting on the maximum value of Q.sub.1 that the UE supports. For example, when the UE reports the support of a maximum value of Q.sub.1 equal to 3. Then, the configured Q.sub.1 value can be from {1,2,3}.
(254) In one example (I.5.5), the beam reporting according to this embodiment is configured (or used) for at least one of the following purposes. In one example (I.5.5.1), for UL multi-beam operations such as UL beam indication (via UL-TCI or joint TCI) indicating a UL Tx beam for UL transmission of UL data and/or control. In one example (I.5.5.2), for both UL and/or UL multi-beam operations such as UL beam indication (via UL-TCI or joint TCI) indicating a UL Tx beam for UL transmission of UL data and/or control and/or DL beam indication (via DL-TCI or joint TCI) indicating a DL Rx beam for DL reception of DL data and/or control. In one example (I.5.5.3), for UL multi-beam operations (cf. example (I.5.5.1)) when the UE detects an event such MPE issue (described in this disclosure).
(255) In one example (I.5.6), the UL channel(s) or resource(s) for the beam report (according to this embodiment) is according to at least one of the following: In one example (I.5.6.1), the layer 1 (L1) channel(s) or resource(s) is configured for the beam report. The beam report can be reported on PUCCH. If the PUCCH transmission that includes the beam report overlaps a PUSCH transmission, the uplink control information (UCI) with the beam report is multiplexed in the PUSCH. Or, the beam report can be included in UCI transmitted on PUSCH, wherein, the PUSCH transmission can be one of: a PUSCH transmission scheduled by an UL grant, or a configured grant PUSCH transmission of Type 1 or of Type 2, or a Msg3 PUSCH transmission for random access procedure Type 1, or a MsgA PUSCH transmission for random access procedure Type 2. In one example (I.5.6.2), the layer 2 (L2) MAC CE channel(s) or resource(s) is configured for the beam report. In one example, the single-entry PHR MAC CE entity for MPE reporting (REF11) is used. The MAC CE entity is augmented/extended to include PHR per reported resource indicator and possibly other metrics for MPE reporting. In one example, the multiple-entry PHE MAC CE for MPE reporting (REF11) is used. The MAC CE entity is modified to include PHR per reported resource indicator instead of or in addition to the serving cell PHR. In one example, a new MAC CE entity is used. In one example, the MAC CE entity is included in a PUSCH transmission, wherein, the PUSCH transmission can be one of: a PUSCH transmission scheduled by an UL grant, or a configured grant PUSCH transmission of Type 1 or of Type 2, or a Msg3 PUSCH transmission for random access procedure Type 1, or a MsgA PUSCH transmission for random access procedure Type 2.
(256) In one embodiment (I.6), the UE is configured a set of joint TCI states and/or UL TCI states (e.g., via RRC and/or MAC CE), and is also configured with a beam reporting that doesn't include any UE-recommendation, and the beam report includes Q.sub.1 TCI state IDs (I) or Q.sub.1 pairs of (I,J)=(TCI state ID, beam metric). The UE determines the beam report based on the measurement of the reference or source RSs included in each TCI state of the set of configured TCI states. The rest of the details of the beam reporting are the same embodiment (I.5) (or other embodiments) except that the resource indicator I is replaced with a TCI state ID.
(257) The details of the virtual PHR (or vPH) reporting in the rest of the disclosure is according to the description of the same in embodiment (I.5).
(258) In one embodiment (I.7), as shown in FIG. 10, a UE is configured (by the NW/gNB) to measure (receive) P.sub.1 DL measurement RS resources (such as CSI-RS or SSB), where P.sub.1≥1, and the details about this measurement are as described in embodiment (I.1).
(259) The UE is further configured (by the NW/gNB) to report a beam report, wherein the beam report includes Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), where Q.sub.1≤P.sub.1, and resource indicator and beam metric are according to the examples described in embodiment (I.1). For example, the resource indicator can be SSBRI or CRI, and the beam

metric can be L1-RSRP or L1-SINR, as defined in Rel. 15/16 NR specification.

(260) In addition to Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), the beam report can also include an information about the 'state' of the beam reporting (e.g., state=MPE condition). The time-domain behavior of this beam measurement and/or beam reporting can be configured as aperiodic (AP), semi-persistent (SP), or periodic (P). The time-domain behavior of the beam measurement RS can be configured as aperiodic (AP), semi-persistent (SP), or periodic (P). The information about the 'state' can be reported via a bitmap or bit-field or bit sequence. Or the information about the 'state' can be reported via a parameter or information element (IE).

(261) The 'state' can be reported jointly with the other components included in the beam report using a joint parameter (or indicator). For example, the 'state' can be reported jointly with at least one of the resource indicators. Or the 'state' can be reported jointly with at least one of the beam metrics. Or the 'state' can be reported jointly with at least one of the pairs (resource indicator, beam metric).

(262) Or the 'state' can be reported separately (independently) from the other components in the beam report using a separate parameter (or indicator). The reporting configuration can be joint (one configuration) or separate (two configurations), one for 'state'(s) and another for the other components in the beam report.

(263) Or the reporting of the 'state' and the other components in the beam report are decoupled, i.e., one reporting for the 'state' and another reporting for the other components in the beam report. The reporting configuration can be joint (one configuration) or separate (two configurations), one for 'state'(s) and another for the other components in the beam report.

(264) In one example, when the UE is equipped with $X > 1$ antenna panels, Q.sub.1=X, and one resource indicator (I) or pair of (I,J)=(resource indicator, beam metric) is reported for each antenna panel. In one example, when the UE is equipped with $X > 1$ antenna panels, $Q.sub.1 \geq X$, and at least one resource indicator (I) or pair of (I,J)=(resource indicator, beam metric) is reported for each antenna panel. The set of P.sub.1 DL measurement RS resources can be partitioned into X subsets, one subset for each antenna panel. Or the P.sub.1 DL measurement RS resources is a super-set encompassing X sets, one set for each antenna panel. In one example, the information about X panels is not provided to the NW/gNB. In one example, the information about X panels is provided to the NW/gNB. For instance, the information about the panel ID can be included/reported implicitly, e.g., from the resource indicators (I). Or the information about the panel ID can be included/reported explicitly, e.g., by including/reporting the panel ID(s) in the beam report.

(265) For $X > 1$ antenna panels at the UE, the 'state' can be reported independently for each panel. Or the 'state' can be reported for one of the X panels (e.g., the panel that has the worst MPE issue), and can also include a corresponding panel ID optionally. Or the 'state' can be reported for all panels that have MPE issue. If there is no MPE issue detected, the UE does not report the 'state', otherwise, the UE reports the 'state' for all panels having the MPE issue and can also report the corresponding panels ID(s) optionally.

(266) For $X > 1$ subsets (or sets) of DL measurement RS resources, the 'state' can be reported independently for each subset (or set). Or the 'state' can be reported for one of the X subsets (or sets), e.g., the subset or the set that has the worst MPE issue and can also include a corresponding subset (or set) ID optionally. Or the 'state' can be reported for all subsets (or sets) that have MPE issue. If there is no MPE issue detected, the UE does not report the 'state', otherwise, the UE reports the 'state' for all subsets (or sets) having the MPE issue and can also report the corresponding subset (or set) ID(s) optionally.

(267) The 'state' can be related to an event of interest (such as MPE), or panel status (or ID), or whether the beam report is for one of or both of DL reception and UL transmission. At least one of the following examples can be used for reporting the information about the state. In example A, the state is regarding the MPE event or MPE mitigation. For example, a 1-bit indication B can be used to indicate one of the two states: (i) MPE detected (or MPE mitigation), and (ii) MPE not detected (not for MPE mitigation). For example, B=1 indicates (i) and B=0 indicates (ii), or vice versa. Alternatively, a parameter p is used to indicate one of the two state: (i) and (ii), via value $p=v0$ and $p=v1$, respectively. In example B, the state is regarding a panel entity when the UE is equipped with multiple UE panels. For example, the state can correspond to an ID e.g., panel ID or set ID or resource ID associated with a panel. In example C, the state is regarding a combination of the MPE event and panel entity. For example, the state can indicate a pair (x,y), where x is either (i) or (ii) in example A, and y is an ID as described in example B. In example D, the state is regarding the recommended/preferred usage of the beam report for one of both of DL reception and UL transmission. At least one of the following examples is used/configured. In one example, the state indicates one of the two possible values $S=\{\text{DL reception, none}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{DL reception, UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{DL reception, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{UL transmission, both DL reception and UL transmission}\}$. In one example, the state indicates one of the three possible values $S=\{\text{DL reception, UL transmission, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, DL reception, UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, DL reception, both DL reception and UL transmission}\}$. In one example, the state indicates one of the two possible values $S=\{\text{none, UL transmission, both DL reception and UL transmission}\}$. In one example, the state indicates one of the three possible values $S=\{\text{none, DL reception, UL transmission, both DL reception and UL transmission}\}$. In example E, the state is regarding a combination of the MPE event and the recommended usage of the beam report for one of both of DL reception and UL transmission. For example, the state can indicate a pair (x,y), where x is either (i) or (ii) in example A, and y takes a value from S, as described in example D. In example F, the state is regarding a combination of the panel entity and the recommended usage of the beam report for one of both of DL reception and UL transmission. For example, the state can indicate a pair (x,y), where x is an ID as described in example B, and y takes a value from S, as described in example D. In example G, the state is regarding a combination of the MPE event, the panel entity, and the recommended usage of the beam report for one of both of DL reception and UL transmission. For example, the state can indicate a triple (x,y,z), where x is either (i) or (ii) in example A, y is an ID as described in example B, and z takes a value from S, as described in example D.

(268) Where 'none' implies no reporting of the recommended/preferred usage from the UE. Or 'none' implies that additional information about the state is not reported in the beam report.

(269) Regarding the beam metric for different values of the 'state', at least one of the following examples is used/configured. In one example, the beam metric can remain the same regardless of the reported value of the 'state'. In one example, the beam metric can be modified depending on the reported value of the 'state'. For example, the modification corresponds to a scaling of the beam metric L1-RSRP or L1-SINR. In one example, the beam metric can be m1 for when the reported value of the 'state' belongs to a set T1, and the beam metric can be m2 for when the reported value of the 'state' belongs to a set T2, where m1 and m2 are two different beam metrics, and T1 and T2 are two sets of possible values of the 'state'. The sets T1 and T2 can be disjoint (i.e., no common element). The (m1,m2) is according to at least one of the following examples: (m1,m2)=(L1-RSRP, scaled L1-RSRP) (m1,m2)=(L1-SINR, scaled SINR) (m1,m2)=(L1-RSRP, L1-SINR) (m1,m2)=(L1-RSRP, UL RSRP) (m1,m2)=(L1-RSRP, virtual PHR) (m1,m2)=(L1-SINR, UL RSRP) (m1,m2)=(L1-

SINR, virtual PHR) In one example, the beam metric can be m1 for when the reported value of the ‘state’ belongs to a set T1, the beam metric can be m2 for when the reported value of the ‘state’ belongs to a set T2, and the beam metric can be m3 for when the reported value of the ‘state’ belongs to a set T3, where m1, m2, and m3 are three different beam metrics, and T1, T2, and T3 are three sets of possible values of the ‘state’. The sets T1, T2, and T3 can be disjoint (i.e., no common element). The (m1,m2,m3) is according to at least one of the following examples: (m1,m2,m3)=(L1-RSRP, scaled L1-RSRP, UL RSRP) (m1,m2,m3)=(L1-RSRP, scaled L1-RSRP, virtual PHR) (m1,m2,m3)=(L1-SINR, scaled L1-SINR, UL RSRP) (m1,m2,m3)=(L1-SINR, scaled L1-SINR, virtual PHR) (m1,m2,m3)=(L1-RSRP, UL RSRP, virtual PHR) (m1,m2,m3)=(L1-SINR, UL RSRP, virtual PHR) m1,m2,m3)=(L1-RSRP, L1-SINR, virtual PHR) (m1,m2,m3)=(L1-RSRP, L1-SINR, UL RSRP) In one example, the beam report can include a second metric (in addition to the beam metric) for some reported value(s) of the ‘state’. That is, when the ‘state’ belongs to a set T1, the beam metric is reported (and the second metric is not reported), and when the ‘state’ belongs to a set T2, both the beam metric and the second metric are reported. Whether the second metric can be reported is determined based on at least one of the following examples. In one example, it is based on a configuration (e.g., via RRC and/or MAC CE and/or DCI). In one example, it is determined and reported by the UE. In this case, a two-part UCI (similar to Rel. 15 NR specification) can be used, wherein the UCI part 1 has an information about whether and/or the number of the second metric being reported in UCI part 2. In one example, it is subject to a UE capability reporting.

(270) In one example, Q.sub.1=1, indicating only one beam (resource indicator and metrics) is reported in the beam report. In this case, the ‘state’ value can be fixed, or reported by the UE, the details as explained above.

(271) In one example, when Q.sub.1>1, there are multiple beams (resource indicators and metrics) reported in the beam report. At least one of the following examples is used/configured regarding the ‘state’. In one example, all Q.sub.1 beams correspond to the same value of the ‘state’. For example, when the ‘state’ is related to the MPE, all Q.sub.1 beams are for (i) for MPE mitigation (taking MPE into account) or (ii) without MPE, i.e., are normal beams without taking MPE into account. A common bit field, IE, or parameter (e.g., 1-bit field or a parameter taking two values) is used for Q.sub.1 beams. In one example, N1 out of Q.sub.1 beams can correspond to one value of the ‘state’ (e.g., taking MPE into account) and remaining N2 beams can correspond to another value of the ‘state’ (e.g., normal beams without taking MPE into account). In one example, both N1 and N2 are fixed. In one example, N1 is fixed and N2 is configured (e.g., RRC and/or MAC CE and/or DCI). In one example, N1 is fixed or configured (e.g., RRC and/or MAC CE and/or DCI), and N2 is reported by UE. A 2-part UCI can be used. For example, the value N2 can be from 0 to Q.sub.1–N1. When N2=0, the UCI part 2 is absent not reported. When N2>1, one beam is reported via UCI part 2. Or UCI part 1 includes report for N1 beams, and a portion of report for N2 beams, and UCI part 2 includes remaining portion of report for N2 beams. In one example, both N1 and N2 reported by UE. A 2-part UCI can be used. For example, reported for a fixed number of beams is reported via UCI part 1, and the remaining via UCI part 2. Or UCI part 1 includes report for a portion of report Q.sub.1 beams, and UCI part 2 includes remaining portion of report for Q.sub.1 beams.

(272) In the rest of disclosure, the ‘state’ is assumed to indicate one of two values, whether MPE issue is detected by the UE or not. The embodiments and examples however are applicable other examples of the ‘state’ described above.

(273) In one sub-embodiment (I.7.1), the content or information conveyed by the ‘state’ is determined (or configured) according to at least one of the following examples.

(274) In one example (I.7.1.1), the ‘state’ indicates one of two values, e.g., whether MPE issue is detected by the UE or not. For example, the information can be conveyed via a 1-bit field, where the field value=0 (or 1) indicates that the MPE issue is not detected, and the field value=1 (or 0) indicates that the MPE issue is detected. When the field value=0 (or 1), e.g., indicating that the MPE issue is not detected, the beam report includes beam indicator(s) (e.g., CRI(s)/SSBRI(s) and beam metric(s) (e.g., L1-RSRP, L1-SINR etc.) When the field value=1 (or 0), e.g., indicating that the MPE issue is detected, the beam report includes beam indicator(s) (e.g., CRI(s)/SSBRI(s) and beam metric(s) taking the event indicated by the field value into account, where a beam metric taking the event indicated by the field value can be at least one of the following: Example (I.7.1.1.1): the same as the beam metric when the field value=0, e.g., L1-RSRP, L1-SINR etc. Example (I.7.1.1.2): a modified beam metric, e.g., the modification corresponds to a scaling of the beam metric L1-RSRP or L1-SINR. Example (I.7.1.1.3): a new beam metric, e.g., UL RSRP, or virtual PHR. In one example, the UL RSRP is defined as $UL\ RSRP = L1-RSRP - P_{sub.DL} + P_{sub.UL}$, where L1-RSRP is the measured DL RSRP, P.sub.DL is the DL transmit power, and P.sub.UL, is the available UL transmit power (or max UL transmit power). In one example, the virtual PHR is defined as a difference between the maximum available transmit power (e.g., P.sub.c,max) and the required transmit power.

(275) In one example (I.7.1.2), the information indicates whether MPE issue is detected by the UE or not. For example, the information can be conveyed via a parameter, where the parameter value=v0 indicates that the MPE issue is not detected, and the parameter value=v1 indicates that the MPE issue is detected.

(276) In one example (I.7.1.3), the information includes a value of the MPE. For example, the information can be conveyed via a M-bit field. When M=1, the 1-bit field indicates one of the two values {m.sub.0, m.sub.1}, where either both values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the other maps to the state “MPE detected and a corresponding (maximum or minimum) MPE value”. When M=2, the 2-bit field indicates one of the four values {m.sub.0, m.sub.1, m.sub.2, m.sub.3}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” In general, the M-bit field indicates one of the 2^{sup}.M values {m.sub.0, m.sub.1, . . . , m.sub.M-1}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” The set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1} can be fixed, or configured, e.g., via RRC and/or MAC CE and/or DCI. Or a maximum value for the MPE is configured, e.g., via RRC and/or MAC CE and/or DCI, and the set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1} is determined based on the configured maximum value. In one example, the set of MP values corresponds to the (maximum or minimum) power headroom (PHR) or Power Management Maximum Power Reduction (P-MPR) or UL duty-cycle values.

(277) In one example (I.7.1.4), the information includes a value of the MPE. For example, the information can be conveyed via a parameter taking a value from a set comprising M values. When M=2, the parameter indicates one of the two values {m.sub.0, m.sub.1}, where either both values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the other maps to the state “MPE detected and a corresponding (maximum or minimum) MPE value”. When M=4, the parameter indicates one of the four values {m.sub.0, m.sub.1, m.sub.2, m.sub.3}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” In general, the parameter indicates one of the M values {m.sub.0, m.sub.1, . . . , m.sub.M-1}, where either all values map to (maximum or minimum) MPE values, or one of them maps to the state “MPE issue not detected”, and the remaining map to the states “MPE detected and a corresponding (maximum or minimum) MPE value.” The set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1}

1} can be fixed, or configured, e.g., via RRC and/or MAC CE and/or DCI. Or a maximum value for the MPE is configured, e.g., via RRC and/or MAC CE and/or DCI, and the set of values {m.sub.0, m.sub.1, . . . , m.sub.M-1} is determined based on the configured maximum value. In one example, the set of MP values corresponds to the (maximum or minimum) power headroom (PHR) or Power Management Maximum Power Reduction (P-MPR) or UL duty-cycle values.

(278) In one sub-embodiment (I.7.2), the reporting of the 'state' is determined (or configured) according to at least one of the following examples.

(279) In one example (I.7.2.1), the 'state' can always be included the beam report (i.e., reported by the UE).

(280) In one example (I.7.2.2), the 'state' can always be included the beam report (i.e., reported by the UE) when a condition is met, wherein, for example, the condition being that the UE is equipped with multiple antenna panels, or that the UE reports this (the UE-recommendation) as part of its UE capability, or that an event of interest can be detected by the UE.

(281) In one example (I.7.2.3), the reporting of the 'state' is configured via RRC and/or MAC CE (and/or triggered via DCI). When configured (or triggered), the UE always reports the 'state', otherwise, the UE does not report it. The UE can also be configured with a maximum value for the MPE. In one example, the configuration and/or triggering is UE-specific, in another example, the configuration and/or triggering is UE-group specific, in another example, the configuration and/or triggering is cell-specific.

(282) In one example (I.7.2.4), the reporting of the 'state' is reported by the UE without any configuration/triggering from the NW (i.e., the UE can initiate such reporting).

(283) In one example (I.7.2.5), the capability of reporting of the 'state' is reported by the UE (e.g., via UE capability reporting or as part of an UL transmission). Depending on the reported UE capability, the NW/gNB can configure (or trigger) the reporting of the 'state' via RRC and/or MAC CE (and/or DCI). The UE reports the 'state' according to the configuration/triggering from the NW/gNB.

(284) In one example (I.7.2.6), the UE decides (has the freedom to decide) whether the 'state' is included in the beam report, i.e., the UE may or may not include the 'state' in the beam report. For instance, such a decision can be made based on the event of interest. In one example, the beam report (with or without the 'state') is reported via a one-part UCI (similar to WB CSI reporting in Rel. 15 NR, the UE can append a fixed number of zeros in order to ensure that the payload remains the same regardless of whether the UE reports the 'state' or not). In one example, the beam report (with or without the 'state') is reported via a two-part UCI, wherein part 1 UCI includes an information whether the UE reports the 'state' or not. If the information indicates the 'state' being reported, it is reported via part 2 UCI.

(285) In one sub-embodiment (I.7.3), in addition to Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric) and 'state', the beam report can also include a UE-recommendation for UL TX beam selection/indication (by the NW/gNB), where the details about the UE-recommendation are as described in impediment I.1. Also, the 'state' and the UE-recommendation can either be reported jointly or separately.

(286) Upon receiving the beam report, the NW/gNB configures/indicates the UL TX beam indication (which can be the same as DL TX beam indication since beam correspondence holds) to the UE for UL transmission, where the beam indication indicates $N \geq 1$ UL TX beam(s). The beam indication can be via DL-TCI or UL-TCI or J-TCI (joint TCI) or other functionally equivalent entity such as SpatialRelationInfo or SRI that is indicated via DCI and/or MAC CE and/or RRC. In one example $N=1$. In one example, $N=2$. Also, the beam indication can include a message (such as ACK or NACK/NULL), where the details about the message are as described in embodiment (I.1) and/or (I.4).

(287) In one sub-embodiment (I.7.4), a UE provides two sets of reports, either with a same report or separately in two different reports, wherein; A first report (sub-report) is a beam report including resource indicator(s) and possibly beam metrics, without reflecting the 'state'. A second report (sub-report) is a beam report including resource indicator(s) and possibly beam metrics, reflecting the 'state'. For example, the reported resource indicator(s) take into account the MPE effect. The beam metrics include the MPE effect. In one example, this behavior is configurable/triggered to the UE via RRC and/or MAC CE and/or DCI. When configured, the UE provides two reports (sub-reports), else it provides one report (the first report). In one example, the configuration and/or triggering is UE-specific, in another example, the configuration and/or triggering is UE-group specific, in another example, the configuration and/or triggering is cell-specific.

(288) In one embodiment (I.8), the UE is configured with a beam reporting that doesn't include any state, i.e., the beam report includes Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), where $Q.sub.1 \leq P.sub.1$ as described in embodiment (I.7) and/or (II.1)/(II.2).

(289) In one embodiment (I.9), the UE is configured with a beam reporting that doesn't include any state, i.e., the beam report includes Q.sub.1 resource indicators (I) or Q.sub.1 pairs of (I,J)=(resource indicator, beam metric), where $Q.sub.1 \leq P.sub.1$ as described in embodiment (I.3) or (II.1)/(II.2).

(290) Any of the above variation embodiments can be utilized independently or in combination with at least one other variation embodiment.

(291) FIG. 18 illustrates a flow chart of a method 1800 for operating a user equipment (UE), as may be performed by a UE such as UE 116, according to embodiments of the present disclosure. The embodiment of the method 1800 illustrated in FIG. 18 is for illustration only. FIG. 18 does not limit the scope of this disclosure to any particular implementation.

(292) As illustrated in FIG. 18, the method 1800 begins at step 1802. In step 1802, the UE (e.g., 111-116 as illustrated in FIG. 1) receives information for a maximum permissible emission (MPE) report.

(293) In step 1804, the UE determines the MPE report including Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$.

(294) In step 1806, the UE includes the MPE report in a PHR report.

(295) In step 1808, the UE transmits, via a MAC CE, the PHR report including the MPE report.

(296) In one embodiment, the information for the MPE report is via radio resource control (RRC) parameter mpe-Reporting-FR2-r17 included in an IE PHR-Config that configures parameters for the PHR report.

(297) In one embodiment, the UE triggers the MPE report when a measured power management power reduction (P-MPR) is equal to or larger than an MPE threshold.

(298) In one embodiment, the MPE threshold is configured via mpe-Reporting-FR2-r17.

(299) In one embodiment, a value of Q is configured via mpe-Reporting-FR2-r17.

(300) In one embodiment, a max value of Q is $P \in \{1, 2, 3, 4\}$ and is reported by the UE via capability reporting.

(301) In one embodiment, the resource indicator is synchronization signal physical broadcast channel block resource indicator (SSBRI) or channel state information reference signal resource indicator (CRI), reported using 6 bits, and the MPE metric is a measured P-MPR, reported using 2 bits from a table

(302) TABLE-US-00012 Reported value Measured quantity value Unit P-MPR_00 $3 \leq PMP-R < 6$ dB P-MPR_01 $6 \leq PMP-R < 9$ dB P-MPR_02 $9 \leq PMP-R < 12$ dB P-MPR_03 $PMP-R \geq 12$ dB

(303) FIG. 19 illustrates a flow chart of another method **1900**, as may be performed by a base station (BS) such as BS **102**, according to embodiments of the present disclosure. The embodiment of the method **1900** illustrated in FIG. **19** is for illustration only. FIG. **19** does not limit the scope of this disclosure to any particular implementation.

(304) As illustrated in FIG. **19**, the method **1900** begins at step **1902**. In step **1902**, the BS (e.g., **101-103** as illustrated in FIG. **1**), generates information for a MPE report.

(305) In step **1904**, the BS transmits the information.

(306) In step **1906**, the BS receives, via a MAC CE, a PHR report, the PHR report including the MPE report, wherein the MPE report includes Q pairs of (resource indicator, MPE metric), where $Q \in \{1, \dots, P\}$.

(307) In one embodiment, the information for the MPE report is via radio resource control (RRC) parameter mpe-Reporting-FR2-r17 included in IE PHR-Config that configures parameters for the PHR report.

(308) In one embodiment, the MPE report is triggered when a measured power management power reduction (P-MPR) is equal to or larger than an MPE threshold.

(309) In one embodiment, the MPE threshold is configured via mpe-Reporting-FR2-r17.

(310) In one embodiment, a value of Q is configured via mpe-Reporting-FR2-r17.

(311) In one embodiment, a max value of Q is $P \in \{1, 2, 3, 4\}$ and is based on capability reporting.

(312) In one embodiment, the resource indicator is synchronization signal physical broadcast channel block resource indicator (SSBRI) or channel state information reference signal resource indicator (CRI), reported using 6 bits, and the MPE metric is a measured P-MPR, reported using 2 bits from a table

(313) TABLE-US-00013 Reported value Measured quantity value Unit P-MPR_00 $3 \leq \text{PMP-R} < 6$ dB P-MPR_01 $6 \leq \text{PMP-R} < 9$ dB P-MPR_02 $9 \leq \text{PMP-R} < 12$ dB P-MPR_03 $\text{PMP-R} \geq 12$ dB

(314) The above flowcharts illustrate example methods that can be implemented in accordance with the principles of the present disclosure and various changes could be made to the methods illustrated in the flowcharts herein. For example, while shown as a series of steps, various steps in each figure could overlap, occur in parallel, occur in a different order, or occur multiple times. In another example, steps may be omitted or replaced by other steps.

(315) Although the present disclosure has been described with an exemplary embodiment, various changes and modifications may be suggested to one skilled in the art. It is intended that the present disclosure encompass such changes and modifications as fall within the scope of the appended claims. None of the description in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claims scope. The scope of patented subject matter is defined by the claims.

Claims

1. A user equipment (UE) comprising: a processor; and a transceiver operably coupled to the processor, the transceiver configured to: receive configuration information for a maximum permissible emission (MPE) report, wherein the configuration information includes a resource pool for a power management-maximum power reduction (P-MPR); and transmit, via medium access control-control element (MAC CE), a power headroom (PHR) report, the MAC CE including a MPE metric corresponding to a P-MPR value and a resource indicator indicating a beam for the MPE metric, wherein the configuration information further includes a value indicating a number of P-MPR values included in the MAC CE, the value is one of 1, 2, 3, or 4.
2. The UE of claim 1, wherein the configuration information for the MPE report is via a radio resource control (RRC) signaling.
3. The UE of claim 1, wherein the MPE report is triggered when the P-MPR value is equal to or larger than a threshold value for the P-MPR.
4. The UE of claim 3, wherein the threshold value for the P-MPR is configured via mpe-Reporting-FR2-r17.
5. The UE of claim 1, wherein the value is configured via mpe-Reporting-FR2-r17.
6. The UE of claim 1, wherein the value is reported by the UE via capability reporting.
7. The UE of claim 1, wherein the resource indicator is synchronization signal/physical broadcast channel block resource indicator (SSBRI) or channel state information reference signal resource indicator (CRI).
8. A method for operating a user equipment (UE), the method comprising: receiving configuration information for a maximum permissible emission (MPE) report, wherein the configuration information includes a resource pool for a power management-maximum power reduction (P-MPR); transmitting, via medium access control-control element (MAC CE), a power headroom (PHR) report, the MAC CE including a MPE metric corresponding to a P-MPR value and a resource indicator indicating a beam for the MPE metric, wherein the configuration information further includes a value indicating a number of P-MPR values included in the MAC CE, the value is one of 1, 2, 3, or 4.
9. The method of claim 8, wherein the configuration information for the MPE report is via a radio resource control (RRC) signaling.
10. The method of claim 8, wherein the MPE report is triggered when the P-MPR value is equal to or larger than a threshold value for the P-MPR.
11. The method of claim 10, wherein the threshold value for the P-MPR is configured via mpe-Reporting-FR2-r17.
12. The method of claim 8, wherein the value is configured via mpe-Reporting-FR2-r17.
13. The method of claim 8, wherein the value is reported by the UE via capability reporting.
14. The method of claim 8, wherein the resource indicator is synchronization signal/physical broadcast channel block resource indicator (SSBRI) or channel state information reference signal resource indicator (CRI).
15. A base station (BS) comprising: a processor; and a transceiver operably coupled to the processor, the transceiver configured to: transmit configuration information for a maximum permissible emission (MPE) report, wherein the configuration information includes a resource pool for a power management-maximum power reduction (P-MPR); and receive, via medium access control-control element (MAC CE), a power headroom (PHR) report, the MAC CE including a MPE metric corresponding to a P-MPR value and a resource indicator indicating a beam for the MPE metric, wherein the configuration information further includes a value indicating a number of P-MPR values included in the MAC CE, the value is one of 1, 2, 3, or 4.
16. The BS of claim 15, wherein the configuration information for the MPE report is via a radio resource control (RRC) signaling.
17. The BS of claim 15, wherein the MPE report is triggered when the P-MPR value is equal to or larger than a threshold value for the P-MPR.

18. The BS of claim 15, wherein the resource indicator is synchronization signal/physical broadcast channel block resource indicator (SSBRI) or channel state information reference signal resource indicator (CRI).
